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MATHEMATICS

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Integral Equation

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INTEGRAL EQUATION

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CHAPTER 1

INTEGRAL EQUATION

1.1. Definition

An integral equation is an equation in which an unknown function, to be determined, appears under one or more integral signs. If the derivatives of the function are involved, it is called an integro-differential equation.

An equation of the form

$$\alpha(x)\phi(x) + F(x) + \lambda \int_{\Omega} K(x, \xi)\phi(\xi)d\xi = 0,$$

is called the linear integral equation, where $\phi(x)$ is the unknown function; $\alpha(x)$, $F(x)$ and the kernel of the integral equation $K(x, \xi)$ are known functions; λ is a non-zero real or complex parameter, and the integration extends over the domain Ω of the auxiliary variable ξ .

Linear integral equations are classified into two basic types:

1.2. Volterra Integral Equation

An integral equation is said to be a Volterra integral equation if the upper limit of integration is a variable, e.g.,

$$\alpha(x)\phi(x) = F(x) + \lambda \int_a^x K(x, \xi)\phi(\xi)d\xi$$

- (i) When $\alpha \equiv 0$, the unknown function ϕ appears only under the integral sign and nowhere else in the equation, then

$$F(x) = \lambda \int_a^x K(x, \xi)\phi(\xi)d\xi, \quad a > -\infty$$

is called the Volterra's integral equation of first kind.

- (ii) When $\alpha \equiv 1$, the equation involves the unknown function ϕ , both inside as well as outside the integral sign, then

$$\phi(x) = F(x) + \lambda \int_a^x K(x, \xi)\phi(\xi)d\xi$$

is called the Volterra's equation of second kind.

- (iii) When $\alpha \equiv 1$, $F(x) = 0$, the equation reduced to

$$\phi(x) = \lambda \int_a^x K(x, \xi)\phi(\xi)d\xi \text{ is called the homogeneous Volterra's integral equation of second kind.}$$

1.3. Fredholm Integral Equations

An integral equation is said to be Fredholm integral equation if the domain of integration Ω is fixed, e.g.,

$$\alpha(x)\phi(x) = F(x) + \lambda \int_a^b K(x, \xi)\phi(\xi)d\xi$$

- (i) When $\alpha \equiv 0$, the equation involves the unknown function ϕ only under the integral sign, then

$$F(x) = \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi, \quad a \leq x \leq b$$

is called the Fredholm integral equation of first kind.

- (ii) When $\alpha \equiv 1$, the equation involves the unknown function ϕ both inside as well as outside the Integral sign, then

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi, \quad a \leq x \leq b$$

is called the non-homogeneous Fredholm integral equation of second kind.

- (iii) When $\alpha \equiv 1$, $F(x) \equiv 0$, the equation reduced to

$$\phi(x) = \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi, \quad a \leq x \leq b$$

is called as the homogeneous Fredholm integral equation of second kind.

1.4. Differentiation of Function Under an Integral Sign

Consider the function $I_n(x)$ defined by the relation

$$I_n(x) = \int_a^x (x-\eta)^{n-1} f(\eta) d\eta, \quad (1)$$

Where η is a positive integral and a is a constant.

We known that

$$\frac{d}{dx} \int_{P(x)}^{Q(x)} F(x, \eta) d\eta = \int_P^Q \frac{\partial}{\partial x} \{F(x, \eta)\} d\eta + F[x, Q(x)] \frac{dQ}{dx} - F[x, P(x)] \frac{dP}{dx},$$

Which is valid if F and $\frac{\partial F}{\partial x}$ are continuous functions of both x, η and the first derivative of $P(x)$

and $Q(x)$ are continuous.

Differentiating (1) under the integral sign, we have

$$\begin{aligned} \frac{dI_n}{dx} &= (n-1) \int_a^x (x-\eta)^{n-2} f(\eta) d\eta \times [(x-\eta)^{n-1} f(\eta)]_{\eta=x} \\ \frac{dI_n}{dx} &= (n-1) I_{n-1}, \quad n > 1 \end{aligned} \quad (2)$$

From the relation (1), we have

$$I_1(x) = \int_a^x f(\eta) d\eta \Rightarrow \frac{dI_1}{dx} = f(x) \quad (3)$$

Differentiating (2) successively m times, we have

$$\frac{d^m I_n}{dx^m} = (n-1)(n-3)\dots(n-m) I_{n-m}, \quad n > m$$

In particular, we have

$$\frac{d^{n-1}I_n}{dx^{n-1}} = (n-1)! I_1(x)$$

$$\Rightarrow \frac{d}{dx} \left(\frac{d^{n-1}I_n}{dx^{n-1}} \right) = (n-1)! \frac{dI_1}{dx} = (n-1)! f(x) \quad (4)$$

Thus, we have

$$I_1(x) = \int_a^x f(x_1) dx_1$$

$$\frac{dI_2}{dx} = I_1 = \int_a^x f(x_1) dx_1$$

$$\Rightarrow I_2(x) = \int_a^x \int_a^{x_2} f(x_1) dx_1 dx_2$$

$$\text{In general, we have } I_n(x) = (n-1)! \int_a^x \int_a^{x_2} \dots \int_a^{x_{n-1}} f(x_1) dx_1 dx_2 \dots dx_{n-1} dx_n \quad (5)$$

From the relations (1) and (5), we have

$$\int_a^x \int_a^{x_2} \dots \int_a^{x_{n-1}} f(x_1) dx_1 dx_2 \dots dx_{n-1} dx_n$$

$$= \frac{1}{(n-1)!} I_n(x) = \frac{1}{(n-1)!} \int_a^x (x-\eta)^{n-1} f(\eta) d\eta$$

This may be represented as the result of integrating the function f from a to x and then integrating $(n-1)$ times, we have

$$\int_a^x f(\eta) d\eta^n = \int_a^x \frac{(x-\eta)^{n-1}}{(n-1)!} f(\eta) d\eta, \quad (6)$$

1.5. Relation between Differential and Integral Equations

There is a fundamental relationship between integral equations and ordinary and partial differential equations with given initial values. Consider the differential equation of n th order as

$$\frac{d^n y}{dx^n} + a_1(x) \frac{d^{n-1} y}{dx^{n-1}} + a_2(x) \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_n(x) y = F(x) \quad (1)$$

with continuous coefficients $a_i(x) = i = 1, 2, 3, \dots, n$. The initial conditions are prescribed as follows:

$$y(0) = C_0, y'(0) = C_1, y''(0) = C_2, \dots, y^{(n-1)}(0) = C_{n-1} \quad (2)$$

Where the prime denotes differentiation with respect to x . Consider

$$\frac{d^n y}{dx^n} = \phi(x) \text{ By integrating and using the initial conditions (2), we have}$$

$$\frac{d^{n-1} y}{dx^{n-1}} = \int_0^x \phi(\xi) d\xi + C_{n-1}$$

$$\frac{d^{n-2} y}{dx^{n-2}} = \int_0^x \phi(\xi) d\xi^2 + C_{n-1}x + C_{n-2}$$

$$y = \int_0^x \phi(\xi) d\xi^n + C_{n-1} \frac{x^{n-1}}{(n-1)!} + C_{n-2} \frac{x^{n-2}}{(n-2)!} \dots + C_0 \quad (3)$$

Where $\int_0^x \phi(\xi) d\xi^n$ represents for a multiple integral of order n.

From the relations (3) and (1), we obtain

$$\begin{aligned} \phi(x) + a_1(x) \int_0^x \phi(\xi) d\xi + a_2 \int_0^x \phi(\xi) d\xi^2 + \dots + a_n(x) \int_0^x \phi(\xi) d\xi^n \\ = F(x) + \sum_{i=1}^n i C_i \chi_i(x) \end{aligned} \quad (4)$$

$$\text{Where } \chi_i(x) = a_i(x) + \frac{x}{1!} a_{i+1}(x) + \dots + a_n(x) \frac{x^{n-i}}{(n-i)!} \quad (5)$$

$$\phi(x) + \int_0^x \left[a_1(x) + a_2(x)(x-\xi) + \dots + a_n(x) \frac{(x-\xi)^{n-1}}{(n-1)!} \right] \phi(\xi) d\xi = G_i(x), \quad (6)$$

$$\text{Where } G_i(x) = F(x) + \sum_{i=1}^n i C_i \chi_i(x)$$

The equation (6) represents the non-homogeneous Volterra's integral equation of second kind.

1.6. Particular Case

Consider the linear differential equation of second order.

$$\frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x) y = F(x), \quad (1)$$

with initial conditions

$$y(0) = C_0 \text{ and } y'(0) = C_1 \quad (2)$$

$$\text{Consider } \frac{d^2 y}{dx^2} = \phi(x)$$

By integrating and using the initial conditions (2), we have

$$\frac{dy}{dx} = \int_0^x \phi(\xi) d\xi + C_1 \quad (3)$$

$$\text{and } y = \int_0^x (x-\xi) \phi(\xi) d\xi + C_1 x + C_0 \quad (4)$$

The given differential equation reduces to

$$\phi(x) + a_1(x) \left[\int_0^x \phi(\xi) d\xi + C_1 \right] + a_2(x) \left[\int_0^x (x-\xi) \phi(\xi) d\xi + C_1 x + C_0 \right] = F(x)$$

$$\left[\int_0^x (x-\xi) \phi(\xi) d\xi + C_1 x + C_0 \right] y = F(x)$$

$$\text{or } \phi(x) + \int_0^x [a_1(x) + a_2(x)(x-\xi)] \phi(\xi) d\xi$$

$$= F(x) - C_1 a_1(x) - C_1 x a_2(x) - C_0 a_2(x)$$

$$\text{or } \phi(x) = f(x) + \lambda \int_0^x K(x, \xi) \phi(\xi) d\xi \quad (5)$$

Where $K(x, \xi) = a_1(x) + a_2(x)(x - \xi)$, $\lambda = -1$

$$f(x) = F(x) - C_1 a_1(x) - C_1 x a_2(x) - C_0 a_2(x) \quad (6)$$

Which represents the Volterra's integral equation of the second kind. Similarly, the boundary value problems in ordinary differential equations lead to Fredholm integral equations.

Example: Show that the function $\phi(x) = (1 + x^2)^{-3/2}$ is a solution of the Volterra integral equation

$$\phi(x) = \frac{1}{1+x^2} - \int_0^x \frac{\xi}{1+x^2} \phi(\xi) d\xi$$

Solution: Substituting the function $\phi(x) = (1 + x^2)^{-3/2}$ in the given equation, we have

$$\begin{aligned} & \frac{1}{1+x^2} - \int_0^x \frac{\xi}{1+x^2} \cdot \frac{1}{(1+\xi^2)^{3/2}} d\xi \\ &= \frac{1}{1+x^2} + \frac{1}{1+x^2} \left\{ \frac{1}{(1+\xi^2)^{1/2}} \right\}_0^x \\ &= \frac{1}{1+x^2} + \frac{1}{(1+x^2)^{3/2}} - \frac{1}{1+x^2} = \frac{1}{(1+x^2)^{3/2}} = \phi(x) \end{aligned}$$

The substitution of $\phi(x)$ reduces the given equation to an identity with respect to x , thus

$\phi(x) = (1 + x^2)^{-3/2}$ is a solution of the integral equation.

Example: From an integral equation corresponding to the differential equation

$$\frac{d^2 y}{dx^2} - \sin x \frac{dy}{dx} + e^x y = x$$

with the initial conditions

$$y(0) = 1, y'(0) = -1$$

$$\text{Solution: Consider } \frac{d^2 y}{dx^2} = \phi(x) \quad (1)$$

$$\text{Then } \frac{dy}{dx} = \int_0^x \phi(\xi) d\xi - 1 \quad (2)$$

$$\text{And } y = \int_0^x (x - \xi) \phi(\xi) d\xi - x + 1 \quad (3)$$

Substituting the relations (1), (2) and (3) in the given differential equation, we have

Put Your Own Notes

$$\phi(x) - \sin x \left[\int_0^x \phi(\xi) d\xi - 1 \right] + e^x \left[\int_0^x (x - \xi) \phi(\xi) d\xi - x + 1 \right] = x$$

$$\Rightarrow \phi(x) = [x - \sin x + e^x(x - 1)] + \int_0^x [\sin x - e^x(x - \xi)] \phi(\xi) d\xi$$

represents a Volterra's integral equation of second kind.

Example: Reduce the initial integral equation of second kind.

$$\phi''(x) + \lambda \phi(x) = F(x),$$

$$\text{with } \phi(0) = 1, \phi'(0) = 0.$$

Solution: The differential equation is given as

$$\phi''(x) + \lambda \phi(x) = F(x)$$

$$\Rightarrow \phi''(x) = F(x) - \lambda \phi(x)$$

Integrating both the sides with regard to x , we have

$$\int_0^x \phi''(x) dx = \int_0^x \{F(x) - \lambda \phi(x)\} dx$$

$$\Rightarrow [\phi'(x)]_0^x = \int_0^x \{F(x) - \lambda \phi(x)\} dx$$

$$\Rightarrow \phi'(x) - \phi'(0) = \int_0^x \{F(x) - \lambda \phi(x)\} dx$$

$$\Rightarrow \phi'(x) = \int_0^x \{F(x) - \lambda \phi(x)\} dx$$

Integrating both the sides with regard to x , we have

$$\int_0^x \phi'(x) dx = \int_0^x \{F(x) - \lambda \phi(x)\} dx^2$$

$$\Rightarrow \phi(x) - \phi(0) = \int_0^x \{F(\xi) - \lambda \phi(\xi)\} d\xi^2$$

$$\Rightarrow \phi(x) = 1 + \int_0^x (x - \xi) \{F(\xi) - \lambda \phi(\xi)\} d\xi,$$

Which reduces to a Volterra's integral equation of second kind.

Example: Reduce the differential equation

$$\phi''(x) - 3\phi'(x) + 2\phi(x) = 4 \sin x$$

with the conditions

$$\phi(0) = 1, \phi'(0) = -2$$

into a non-homogeneous Volterra's integral equation of second kind. Conversely, derive the original differential equation with the initial conditions from the integral equation obtained.

Solution: The given differential equation may be written as

$$\phi''(x) = 4 \sin x - 2\phi(x) + 3\phi'(x) \quad (1)$$

integrating with regard to x , both the sides we have

$$\int_0^x \phi''(x) dx = 4 \int_0^x \sin x dx - 2 \int_0^x \phi(x) dx + 3 \int_0^x \phi'(x) dx$$

$$\begin{aligned}\Rightarrow [\phi'(x)]_0^x &= -4(\cos x)_0^x - 2 \int_0^x \phi(x) dx + 3 [\phi(x)]_0^x \\ \Rightarrow \phi'(x) - \phi'(0) &= -4(\cos x - 1) - 2 \int_0^x \phi(x) dx + 3 \{\phi(x) - 1\} \\ \Rightarrow \phi'(x) &= -2 - 4(\cos x - 1) + 3[\phi(x) - 1] - 2 \int_0^x \phi(x) dx \\ \Rightarrow \phi'(x) &= -1 - 4 \cos x + 3\phi(x) - 2 \int_0^x \phi(x) dx \quad (1)\end{aligned}$$

integrating with regard to x , both the sides, we have

$$\begin{aligned}\int_0^x \phi'(x) dx &= -\int_0^x dx - 4 \int_0^x \cos x dx + 3 \int_0^x \phi(x) dx - 2 \int_0^x \phi(x) dx^2 \\ \Rightarrow [\phi(x)]_0^x &= -x - 4 \sin x + 3 \int_0^x \phi(x) dx - 2 \int_0^x \phi(x) dx^2 \\ \Rightarrow \phi(x) - \phi(0) &= -x - 4 \sin x + 3 \int_0^x \phi(\xi) d\xi - 2 \int_0^x \phi(\xi) d\xi^2 \\ \Rightarrow \phi(x) - 1 &= -x - 4 \sin x + 3 \int_0^x \phi(\xi) d\xi - 2 \int_0^x (x - \xi) \phi(\xi) d\xi \\ \Rightarrow \phi(x) &= (1 - x - 4 \sin x) + \int_0^x [3 - 2(x - \xi)] \phi(\xi) d\xi \quad (3)\end{aligned}$$

Represents the non-homogeneous Volterra's integral equation of second kind.

Converse: Again, differentiating the equation (3) with regard to x , we have

$$\begin{aligned}\phi'(x) &= -1 - 4 \cos x + \frac{d}{dx} \int_0^x [3 - 2(x - \xi)] \phi(\xi) d\xi \\ \Rightarrow \phi'(x) &= -1 - 4 \cos x + \int_0^x \frac{\partial}{\partial x} [3 - 2(x - \xi)] \phi(\xi) d\xi + [3 - 2(x - x)] \phi(x) \cdot 1 \\ \Rightarrow \phi'(x) &= -1 - 4 \cos x + 3\phi(x) - 2 \int_0^x \phi(\xi) d\xi \quad (4)\end{aligned}$$

Differentiating both the sides with regard to x , we have

$$\begin{aligned}\Rightarrow \phi''(x) &= 4 \sin x + 3\phi'(x) - 2 \frac{d}{dx} \int_0^x \phi(\xi) d\xi \\ \Rightarrow \phi''(x) &= 4 \sin x + 3\phi'(x) - 2\phi(x) \\ \Rightarrow \phi''(x) - 3\phi'(x) + 2\phi(x) &= 4 \sin x\end{aligned}$$

Which is the required given differential equation. Putting $x=0$ in the equations (3) and (4), we

have $\phi(0)=1, \phi'(0)=-2$.

Example: Convert the differential equation $\frac{d^2\phi}{dx^2} + \lambda\phi = 0$

With initial conditions $\phi(0)=0, \phi(l)=0$, into Fredholm integral equation of second kind. Also, recover the original differential equation from the integral equation you obtain.

Solution: The differential equation may be written as

$$\frac{d^2\phi}{dx^2} = -\lambda\phi \quad (1)$$

Integrating both the sides with regard to x , we have

$$\int_0^x \frac{d^2\phi}{dx^2} dx = -\lambda \int_0^x \phi dx$$

$$\left(\frac{d\phi}{dx}\right)_0^x = -\lambda \int_0^x \phi dx$$

$$\phi'(x) - \phi'(0) = -\lambda \int_0^x \phi dx$$

Consider $\phi'(0) = C$, a constant, then $\phi'(x) = C - \lambda \int_0^x \phi(x) dx$

Integrating both the sides, we have

$$[\phi(x)]_0^x = Cx - \lambda \int_0^x \phi(x) dx^2$$

$$\Rightarrow \phi(x) - \phi(0) = Cx - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi$$

$$\Rightarrow \phi(x) = Cx - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi \quad (2)$$

Since $\phi(l) = 0 \Rightarrow \phi(l) = Cl - \lambda \int_0^l (l-\xi)\phi(\xi) d\xi$

$$\Rightarrow 0 = Cl - \lambda \int_0^l (l-\xi)\phi(\xi) d\xi$$

$$\Rightarrow C = \frac{\lambda}{l} \int_0^l (l-\xi)\phi(\xi) d\xi \quad (3)$$

From the relations (2) and (3), we have

$$\phi(x) = \frac{\lambda}{l} x \int_0^l (l-\xi)\phi(\xi) d\xi - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi$$

$$\Rightarrow \phi(x) = \int_0^x \frac{\lambda x(l-\xi)}{l} \phi(\xi) d\xi + \int_x^l \frac{\lambda x(l-\xi)}{l} \phi(\xi) d\xi - \int_0^x \lambda(x-\xi)\phi(\xi) d\xi$$

$$\Rightarrow \phi(x) = \int_0^x \left[\frac{\lambda x(l-\xi)}{l} - \lambda(x-\xi) \right] \phi(\xi) d\xi + \int_x^l \left[\frac{\lambda x(l-\xi)}{l} \phi(\xi) \right]$$

$$\Rightarrow \phi(x) = \lambda \int_0^x \frac{\xi(1-\xi)}{l} \phi(\xi) d\xi + \int_x^l \frac{x(l-\xi)}{l} \phi(\xi) \cdot d\xi \Bigg\}$$

$$\Rightarrow \phi(x) = \lambda \int_0^l K(x, \xi) \phi(\xi) d\xi, \quad (4)$$

$$K(x, \xi) = \begin{cases} \frac{\xi(l-\xi)}{l} & \text{if } 0 \leq \xi \leq x \\ \frac{x(l-\xi)}{l} & \text{if } x \leq \xi \leq l \end{cases} \quad (5)$$

This determines the homogeneous Fredholm integral equation of second kind.

Converse: The integral equation may be taken as

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$$\phi(x) = \int_0^l \frac{\lambda x(l-\xi)}{l} \phi(\xi) d\xi - \int_0^x \lambda(x-\xi) \phi(\xi) d\xi \quad (6)$$

Differentiating both the sides with regard to x , we have

$$\begin{aligned} \phi'(x) &= \frac{d}{dx} \int_0^l \frac{\lambda x(l-\xi)}{l} \phi(\xi) d\xi - \frac{d}{dx} \int_0^x \lambda(x-\xi) \phi(\xi) d\xi \\ \Rightarrow \phi'(x) &= \int_0^l \frac{\partial}{\partial x} \left\{ \frac{\lambda x(l-\xi)}{l} \phi(\xi) \right\} d\xi - \int_0^x \frac{\partial}{\partial x} \{ \lambda(x-\xi) \} \phi(\xi) d\xi \\ \Rightarrow \phi'(x) &= \int_0^l \frac{\lambda(l-\xi)}{l} \phi(\xi) d\xi - \int_0^x \lambda \phi(\xi) d\xi \end{aligned} \quad (7)$$

again differentiating both sides with regard to x , we have

$$\begin{aligned} \Rightarrow \phi''(x) &= \frac{d}{dx} \int_0^l \frac{\lambda(l-\xi)}{l} \phi(\xi) d\xi - \frac{d}{dx} \int_0^x \lambda \phi(\xi) d\xi \\ \Rightarrow \phi''(x) &= \lambda \int_0^l \frac{\partial}{\partial x} \left\{ \frac{(l-\xi)}{l} \phi(\xi) \right\} d\xi - \int_0^x \frac{\partial}{\partial x} [\lambda \phi(\xi)] d\xi - \lambda \phi(x) \\ \Rightarrow \phi''(x) + \lambda \phi(x) &= 0 \end{aligned}$$

From the relation (6), we have

$$\phi(0) = 0 \text{ and } \phi(l) = \int_0^l \left\{ \frac{\lambda l(l-\xi)}{l} - \lambda(l-\xi) \right\} \phi(\xi) d\xi = 0 \quad (8)$$

Which is the required given differential equation.

Example: Obtain Fredholm integral equation of second kind corresponding to the boundary

value problems.

$$\frac{d^2 \phi}{dx^2} = x - \lambda \phi; \phi(0) = 0, \phi'(1) = 0$$

Also, recover the boundary value problem from the integral equation you obtain.

Solution: The differential equation may be written as

$$\frac{d^2 \phi}{dx^2} = x - \lambda \phi$$

Integrating, both the sides, with regard to x , we have

$$\begin{aligned} \int_0^x \frac{d^2 \phi}{dx^2} dx &= \int_0^x x dx - \lambda \int_0^x \phi(x) dx \\ \Rightarrow \phi'(x) - \phi'(0) &= \frac{1}{2} x^2 - \lambda \int_0^x \phi(\xi) d\xi \end{aligned}$$

Integrating both the sides with regard to x , we have

$$\phi(x) - \phi(0) = Cx + \left(\frac{1}{6} \right) x^3 - \lambda \int_0^x \phi(\xi) d\xi^2$$

Put Your Own Notes

$$\Rightarrow \phi(x) = Cx + \left(\frac{1}{6}\right)x^3 - \lambda \int_0^x \phi(\xi) d\xi^2$$

$$\Rightarrow \phi(x) = Cx + \left(\frac{1}{6}\right)x^3 - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi$$

$$\text{Since } \phi'(1) = 0 \Rightarrow \phi'(1) = C + \frac{1}{2} - \lambda \int_0^1 \phi(\xi) d\xi$$

$$\Rightarrow C = -\frac{1}{2} + \lambda \int_0^1 \phi(\xi) d\xi$$

$$\Rightarrow \phi(x) = -\left(\frac{1}{2}\right)x + \left(\frac{1}{6}\right)x^3 + \lambda \int_0^1 \phi(\xi) d\xi - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi$$

$$\Rightarrow \phi(x) = -\left(\frac{1}{2}\right)x + \left(\frac{1}{6}\right)x^3 + \lambda \left\{ \int_0^x x\phi(\xi) d\xi + \int_0^1 x\phi(\xi) d\xi \right\} - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi$$

$$\Rightarrow \phi(x) = \left(\frac{1}{6}\right)(x^3 - 3x) + \lambda \left\{ \int_0^x \xi\phi(\xi) d\xi + \int_x^1 x\phi(\xi) d\xi \right\}$$

$$\phi(x) = \left(\frac{1}{6}\right)(x^3 - 3x) + \lambda \int_0^1 K(x, \xi)\phi(\xi) d\xi \quad (1)$$

$$\text{Where } K(x, \xi) = \begin{cases} x, & x < \xi \\ \xi, & x > \xi \end{cases}$$

This determined the non-homogeneous Fredholm integral equation of second. Put $l=1$ everywhere.

Converse: The integral equation may be taken as

$$\phi(x) = -\left(\frac{1}{2}\right)x + \left(\frac{1}{6}\right)x^3 + \lambda \int_0^1 x\phi(\xi) d\xi - \lambda \int_0^x (x-\xi)\phi(\xi) d\xi \quad (2)$$

Differentiating (2) with regard to x , both the sides, we have

$$\phi'(x) = -\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)x^2 + \lambda \frac{d}{dx} \int_0^1 x\phi(\xi) d\xi - \lambda \frac{d}{dx} \int_0^x (x-\xi)\phi(\xi) d\xi$$

$$\Rightarrow \phi'(x) = -\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)x^2 + \lambda \int_0^1 \phi(\xi) d\xi - \lambda \int_0^x \phi(\xi) d\xi \quad (3)$$

Differentiating again with regard to x , both the sides, we have

$$\phi''(x) = x + \lambda \frac{d}{dx} \int_0^1 \phi(\xi) d\xi - \lambda \frac{d}{dx} \int_0^x \phi(\xi) d\xi$$

$$\Rightarrow \phi''(x) = x + \lambda \int_0^1 \frac{\partial}{\partial x} \{\phi(\xi)\} d\xi - \lambda \int_0^x \frac{\partial}{\partial x} \{\phi(\xi)\} d\xi - \lambda\phi(x)$$

$$\Rightarrow \phi''(x) = x - \lambda\phi(x)$$

$$\Rightarrow \phi''(x) + \lambda\phi(x) = x \quad (4)$$

From the relations (2) and (3), we have

$$\phi(0) = 0 \text{ and } \phi'(1) = -\frac{1}{2} + \frac{1}{2} = 0 \quad (5)$$

Which is the required differential equation together with the boundary conditions.

Put Your Own Notes

EXERCISE 1

- Form an integral equation corresponding to the differential equation.

$$\frac{d^2 y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0 \text{ with the initial conditions}$$

$$y(0) = 0, y'(0) = -1$$

- From the integral equations corresponding to the following differential equations with given initial conditions.

$$(a) \frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 5y = 0; y''(0) = c_1, y'(0) = c_2, y(0) = c_3$$

$$(b) \frac{d^2 y}{dx^2} + (1 + x^2)y = \cos x; y(0) = 0, y'(0) = 2$$

$$(c) \phi'' - \sin x \phi' + e^x \phi = x, \phi(0) = 1, \phi'(0) = -1$$

$$(d) \frac{d^2 \phi}{dx^2} + \lambda \phi = 0; \phi(0) = 0, \phi(l) = 0$$

CHAPTER 2

VOLTERRA'S INTEGRAL EQUATION

2.1. Definition

Solution of non-homogeneous Volterra's integral equation of second kind by the method of successive substitution

Consider the Volterra's integral equation of second kind as

$$\phi(x) = F(x) + \lambda \int_a^x K(x, \xi) \phi(\xi) d\xi, \quad (1)$$

Where

- (i) The kernel $K(x, \xi) \neq 0$ is real and continuous in the rectangle $R : a \leq x \leq b, a \leq \xi \leq b$. Consider $|K(x, \xi)| \leq P$, where P is the maximum value in R .
- (ii) The function $F(x) \neq 0$ is real and continuous in an interval $a \leq x \leq b$. Consider $|F(x)| \leq Q$, where Q is the maximum value in the interval.
- (iii) λ is a non-zero numerical parameter.

Substituting the unknown function $\phi(\xi)$ under an integral sign from the equation (1) itself, we have

$$\phi(x) = F(x) + \lambda \int_a^x K(x, \xi) \left[F(\xi) + \lambda \int_a^\xi K(\xi, \xi_1) \phi(\xi_1) d\xi_1 \right] d\xi$$

$$\phi(x) = F(x) + \lambda \int_a^x K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^x K(x, \xi) \int_a^\xi K(\xi, \xi_1) \phi(\xi_1) d\xi_1 d\xi$$

Performing the operation successively for $\phi(\xi)$, we have

$$\begin{aligned} \phi(x) &= F(x) + \lambda \int_a^x K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^x K(x, \xi) \int_a^\xi K(\xi, \xi_1) \left[F(\xi_1) + \lambda \int_a^{\xi_1} K(\xi_1, \xi_2) \phi(\xi_2) d\xi_2 \right] d\xi_1 d\xi \\ \Rightarrow \phi(x) &= F(x) + \lambda \int_a^x K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^x K(x, \xi) \int_a^\xi K(\xi, \xi_1) F(\xi_1) d\xi_1 d\xi \\ &\quad + \lambda^3 \int_a^x K(x, \xi) \int_a^\xi K(\xi, \xi_1) \int_a^{\xi_1} K(\xi_1, \xi_2) \phi(\xi_2) d\xi_2 d\xi_1 d\xi + \dots \end{aligned}$$

In general, we have

$$\begin{aligned} \phi(x) &= F(x) + \lambda \int_a^x K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^x K(x, \xi) \int_a^\xi K(\xi, \xi_1) F(\xi_1) d\xi_1 d\xi \\ &\quad + \lambda^3 \int_a^x K(x, \xi) \int_a^\xi K(\xi, \xi_1) \int_a^{\xi_1} K(\xi_1, \xi_2) F(\xi_2) d\xi_2 d\xi_1 d\xi + \dots \end{aligned}$$

$$\int_a^x K(\xi, \xi_1) \int_a^{\xi_1} K(\xi_1, \xi_2) \phi(\xi_2) d\xi_2 d\xi_1 + \dots + \lambda^n \int_a^x K(x, \xi) \int_a^{\xi} K(\xi, \xi_1) \int_a^{\xi_1} K(\xi_1, \xi_2) \dots \int_a^{\xi_{n-2}} K(\xi_{n-2}, \xi_{n-1}) F(\xi_{n-1}) d\xi_{n-1} \dots d\xi_1 d\xi + \lambda^{n+1} \int_a^x K(x, \xi) \int_a^{\xi} K(\xi, \xi_1) \int_a^{\xi_1} K(\xi_1, \xi_2) \dots \int_a^{\xi_{n-1}} K(\xi_{n-1}, \xi_n) \phi(\xi_n) d\xi_n \dots d\xi_1 d\xi \quad (2)$$

Consider the infinite series

$$\phi(x) = F(x) + \lambda \int_a^x K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^x K(x, \xi) \int_a^{\xi} K(\xi, \xi_1) F(\xi_1) d\xi_1 d\xi + \lambda^3 \int_a^x K(x, \xi) \int_a^{\xi} K(\xi, \xi_1) \int_a^{\xi_1} K(\xi_1, \xi_2) F(\xi_2) d\xi_2 d\xi_1 d\xi + \dots \quad (3)$$

$$\text{Let } S_n(x) = \lambda^n \int_a^x K(x, \xi) \dots \int_a^{\xi_{n-2}} K(\xi_{n-2}, \xi_{n-1}) F(\xi_{n-1}) d\xi_{n-1} \dots d\xi_1 d\xi$$

$$|S_n(x)| = |\lambda^n| \left| \int_a^x K(x, \xi) \int_a^{\xi} K(\xi, \xi_1) \dots \int_a^{\xi_{n-2}} K(\xi_{n-2}, \xi_{n-1}) F(\xi_{n-1}) d\xi_{n-1} \dots d\xi_1 d\xi \right|$$

Since $|K(x, \xi)| \leq P$ and $|F(x)| \leq Q$

$$\text{Then } |S_n(x)| \leq |\lambda^n| Q P^n \frac{(x-a)^n}{n!}$$

$$|S_n(x)| \leq |\lambda^n| Q \frac{[P(b-a)]^n}{n!}$$

It follows that the series is convergent for all values of λ , P , $Q(b-a)$ and hence the series (3) is absolutely and uniformly convergent.

$$\text{Again } |S_{n+1}(x)| \leq |\lambda^{n+1}| M P^{n+1} \frac{(x-a)^{n+1}}{(n+1)!}$$

$$|S_{n+1}(x)| \leq |\lambda^{n+1}| M \frac{[P(b-a)]^{n+1}}{(n+1)!}$$

Where M is the maximum value of the absolute value of the function $\phi(x)$ which is continuous in the interval I .

$$a \leq x \leq b$$

$$\lim_{x \rightarrow \infty} S_{n+1}(x) = 0$$

We notice that the function $\phi(x)$ which satisfies the relation (2) is the continuous function given by the infinite series (3). The integral equation (1) has a unique continuous solution in the interval I .

2.2. Solution of non-homogeneous Volterra's integral equation of second kind.

A volterra integral equation of second kind

$\phi(x) = F(x) + \lambda \int_0^x K(x, \xi) \phi(\xi) d\xi$ has one and only one solution, given by the relation

$$\phi(x) = F(x) + \lambda \int_0^x R(x, \xi; \lambda) F(\xi) d\xi$$

Where the resolvent kernel $R(x, \xi; \lambda)$ is the sum of the series

$$R(x, \xi; \lambda) = K(x, \xi) + \sum_{v=1}^{\infty} \lambda^v K_v(x, \xi), \text{ convergent for all values of } \lambda.$$

Consider the Volterra integral equation of second kind

$$\phi(x) = F(x) + \lambda \int_0^x K(x, \xi) \phi(\xi) d\xi \quad (1)$$

Where the kernel $K(x, \xi)$ is a continuous function for $0 \leq x \leq a, 0 \leq \xi \leq a$, and the function $F(x)$ is continuous for $0 \leq x \leq a$.

Consider an infinite power series in ascending powers of λ as

$$\phi(x) = \phi_0(x) + \lambda \phi_1(x) + \lambda^2 \phi_2(x) + \dots + \lambda^n \phi_n(x) + \dots \quad (2)$$

Let the series (2) is a solution of the integral equation (1), then

$$\begin{aligned} & \phi_0(x) + \lambda \phi_1(x) + \lambda^2 \phi_2(x) + \dots + \dots + \lambda^n \phi_n(x) + \dots \\ &= F(x) + \lambda \int_0^x K(x, \xi) [\phi_0(\xi) + \lambda \phi_1(\xi) + \lambda^2 \phi_2(\xi) + \dots + \lambda^n \phi_n(\xi)] d\xi \quad (3) \end{aligned}$$

Equating the coefficients of like power of λ , we get

$$\phi_0(x) = F(x)$$

$$\phi_1(x) = \int_0^x K(x, \xi) \phi_0(\xi) d\xi$$

$$\phi_2(x) = \int_0^x K(x, \xi) \phi_1(\xi) d\xi$$

.....

.....

$$\phi_n(x) = \int_0^x K(x, \xi) \phi_{n-1}(\xi) d\xi \quad (4)$$

.....

Thus it yields a method for a successive approximation of the function $\phi_n(x)$. It may be shown that the series (2) converges uniformly in x and λ , for any λ and $x \in [0, a]$, under these assumptions with regard to $F(x)$ and $K(x, \xi)$, its sum is a unique solution of the equation (1).

Put Your Own Notes

Further, from (4), it follows that

$$\phi_1(x) = \int_0^x K(x, \xi) F(\xi) d\xi$$

$$\phi_2(x) = \int_0^x K(x, \xi) \left\{ \int_0^\xi K(\xi, \xi_1) F(\xi_1) d\xi_1 \right\} d\xi$$

Here $\xi_1 = 0, \xi_1 = \xi; \xi = 0, \xi = x$

By interchanging the order of integration, we have

$$\phi_2(x) = \int_0^x F(\xi_1) d\xi_1 \left\{ \int_{\xi_1}^x K(x, \xi) K(\xi, \xi_1) d\xi \right\}$$

$$\Rightarrow \phi_2(x) = \int_0^x K_2(x, \xi_1) F(\xi_1) d\xi_1 \quad (5)$$

$$\text{Where, } K_2(x, \xi_1) = \int_{\xi_1}^x K(x, \xi) K(\xi, \xi_1) d\xi \quad (6)$$

In general, we have

$$\phi_n(x) = \int_0^x K_n(x, \xi) F(\xi) d\xi, \quad n=1, 2, \dots \quad (7)$$

The functions $K_n(x, \xi)$ are called iterated kernels, which can readily be shown that

$$K_1(x, \xi) = K(x, \xi)$$

and $K_2(x, \xi) = K_3(x, \xi)$ etc., are defined recursively by the formulas

$$K_{n+1}(x, \xi) = \int_\xi^x K(x, z) K_n(z, \xi) dz, \quad n=1, 2, \dots \quad (8)$$

The relation (2), which represents the solution of the integral equation, (1) can therefore be written as

$$\phi(x) = F(x) + \sum_{v=1}^{\infty} \lambda^{v-1} \int_0^x K_v(x, \xi) F(\xi) d\xi \quad (9)$$

Where

$$R(x, \xi, \lambda) = K_1(x, \xi) + \lambda K_2(x, \xi) + \lambda^2 K_3(x, \xi) + \dots + \lambda^{n+1} K_n(x, \xi) + \lambda^n K_{n+1}(x, \xi) + \dots \quad (10)$$

$$\Rightarrow R(x, \xi, \lambda) = \sum_{v=0}^{\infty} \lambda^v \int_0^x K_{v+1}(x, \xi) d\xi$$

The function $R(x, \xi, \lambda)$ is called the resolvent kernel or reciprocal kernel of the integral equation (1). The series converges absolutely and uniformly in the case of a continuous kernel $K(x, \xi)$.

Iterated kernels and the resolvent kernel do not depend on the lower limit of an integral equation. The resolvent kernel $R(x, \xi, \lambda)$ satisfies the functional equation.

$$R(x, \xi, \lambda) = K(x, \xi) + \lambda \int_\xi^x K(x, z) R(z, \xi; \lambda) dz$$

Thus the solution of the equation (1) reduces to

$$\phi(x) = F(x) + \lambda \int_0^x R(x, \xi; \lambda) F(\xi) d\xi$$

Example: 1 With the aid of the resolvent kernel find the solution of the integral equation

$$(a) \phi(x) = x + \int_0^x (\xi - x) \phi(\xi) d\xi$$

Solution: (i) Here $K_1(x, \xi) = K(x, \xi) = \xi - x$ (1)

and $K_v(x, \xi) = \int_\xi^x K(x, z) K_{v-1}(z, \xi) dz$ (2)

Substituting $v = 2, 3, 4, \dots$ in the relation (2), we have

$$K_2(x, \xi) = \int_\xi^x K(x, z) K_1(z, \xi) dz$$

$$\Rightarrow K_2(x, \xi) = \int_\xi^x (z - x)(\xi - z) dz = -\frac{1}{3!}(\xi - x)^3$$

or $K_3(x, \xi) = \int_\xi^x K(x, z) K_2(z, \xi) dz$

$$\Rightarrow K_3(x, \xi) = \int_\xi^x (z - x) \left\{ -\frac{1}{3!}(\xi - z)^3 \right\} dz = -\frac{1}{5!}(\xi - x)^5$$

By mathematical induction, we have

$$K_v(x, \xi) = (-1)^{v-1} \frac{1}{(2v-1)!} (\xi - x)^{2v-1}, \quad v = 1, 2, 3, \dots$$

The resolvent kernel is defined as

$$R(x, \xi, \lambda) = \sum_{v=1}^{\infty} \lambda^{v-1} K_v(x, \xi) = \sum_{v=1}^{\infty} K_v(x, \xi) \lambda^{v-1}$$

$$\Rightarrow R(x, \xi, \lambda) = \frac{(\xi - x)}{1!} - \frac{(\xi - x)^3}{3!} + \frac{(\xi - x)^5}{5!} - \dots$$

$$\Rightarrow R(x, \xi, \lambda) = \sin(\xi - x)$$

The solution of the integral equation is determined as

$$\phi(x) = f(x) + \lambda \int_0^x R(x, \xi, \lambda) f(\xi) d\xi$$

$$\Rightarrow \phi(x) = x + \int_0^x \xi \sin(\xi - x) d\xi$$

Example: 2 Solve the integral equation $\phi(x) = 1 + \int_0^x \phi(\xi) d\xi$

Solution: Assume $\phi(x) = \phi_0(x) + \lambda \phi_1(x) + \lambda^2 \phi_2(x) + \dots + \lambda^n \phi_n(x) + \dots$

Here $f(x) = 1$, $K(x, \xi) = 1$, $\lambda = 1$

Equating the coefficients of like powers of λ , we have

$$\phi_0(x) = 1$$

$$\phi_1(x) = \int_0^x d\xi = x$$

$$\phi_2(x) = \int_0^x \xi d\xi = \frac{1}{2} x^2$$

$$\phi_3(x) = \frac{1}{2} \int_0^x \xi^2 d\xi = \frac{1}{3!} x^3$$

and so on.

Thus the solution of the integral equation is given by

$$\phi(x) = 1 + x + \frac{1}{2!} x^2 + \frac{1}{3!} x^3 + \dots = e^x$$

2.3 Determination of Resolvent Kernels

- (a) Consider that the kernel $K(x, \xi)$ is a polynomial of degree $(n-1)$ in ξ such that it may be expressed in the form.

$$K(x, \xi) = a_0(x) + a_1(x)(x-\xi) + a_2(x) \frac{1}{2!} (x-\xi)^2 + \dots + \frac{a_{n-1}(x)}{(n-1)!} (x-\xi)^{n-1} + \dots \quad (1)$$

Where the coefficients $\sum_{v=0}^{n-1} a_v(x)$ are continuous in the integral $[0, a]$.

Let the auxiliary function be

$$\phi(x, \xi, \lambda) = \frac{1}{(n-1)!} (x-\xi)^{n-1} + \lambda \int_{\xi}^x R(t, \xi, \lambda) \frac{(x-t)^{n-1}}{(n-1)!} dt \quad (2)$$

with the conditions

$$\phi = \frac{d\phi}{dx} = \dots = \frac{d^{n-2}\phi}{dx^{n-2}} = 0 \text{ at } x = \xi \text{ and } \frac{d^{n-1}\phi}{dx^{n-1}} = 1 \text{ at } x = \xi \quad (3)$$

In addition, we have

$$R(x, \xi, \lambda) = \frac{1}{\lambda} \frac{d^n}{dx^n} \phi(x, \xi, \lambda) \quad (4)$$

Since the resolvent kernel satisfies the functional equation

$$R(x, \xi, \lambda) = K(x, \xi) + \lambda \int_{\xi}^x K(x, z) R(z, \xi, \lambda) dz \quad (5)$$

From (4) and (5), we have

$$\frac{d^n}{dx^n} \phi(x, \xi, \lambda) = \lambda K(x, \xi) + \lambda \int_{\xi}^x K(x, z) \frac{d^n}{dz^n} \phi(z, \xi, \lambda) dz \quad (6)$$

$$\frac{d^n}{dx^n} \phi(x, \xi, \lambda) = \lambda K(x, \xi) + \lambda \left[K(x, z) \frac{d^{n-1}\phi}{dz^{n-1}} - \frac{\partial K(x, z)}{\partial z} \frac{d^{n-2}\phi}{dz^{n-2}} + \dots + \frac{d^{n-1}K}{dz^{n-1}} \phi \right]_{z=\xi}^x \quad (7)$$

Using (1) and (3) the relation (7) reduces to

$$D\phi = \frac{d^n \phi}{dx^n} - \lambda \left[a_0(x) \frac{d^{n-1}\phi}{dx^{n-1}} + a_1(x) \frac{d^{n-2}\phi}{dx^{n-2}} + \dots + a_{n-1}(x) \phi \right] = 0 \quad (8)$$

The function $\phi(x, \xi; \lambda)$ is therefore the integral of the linear equation $D\phi = 0$ which satisfies the Cauchy conditions.

Thus, we have an expression for the resolvent kernel as

$$R(x, \xi; \lambda) = \frac{1}{\lambda} \frac{d^n}{dx^n} \phi(x, \xi; \lambda) \quad (9)$$

- (b) Further, assume that the kernel $K(x, \xi)$ is a polynomial of degree $(n-1)$ in x such that it may be expressed in the form.

$$K(x, \xi) = b_0(\xi) + b_1(\xi)(\xi - x) + b_2(\xi)\frac{1}{2!}(\xi - x)^2 + \dots + \frac{b_{n-1}(\xi)}{(n-1)!}(\xi - x)^{n-1} + \dots \quad (10)$$

Where the coefficients $b_v(\xi)$ are continuous in the interval $[0, a]$

$$\text{Consider } R(x, \xi; \lambda) = -\frac{1}{\lambda} \frac{d^n}{dx^n} \phi(x, \xi; \lambda) \quad (11)$$

The auxiliary function $\phi(x, \xi; \lambda)$ satisfies the following conditions.

$$\phi = \frac{d\phi}{d\xi} = \dots = \frac{d^{n-2}\phi}{d\xi^{n-2}} = 0 \text{ at } \xi = x; \quad \frac{d^{n-1}\phi}{d\xi^{n-1}} = 1 \text{ at } \xi = x. \quad (12)$$

Therefore the functional relation reduces to

$$\frac{d^n \phi}{d\xi^n} = -\lambda K(x, \xi) + \lambda \int_{\xi}^x K(z, \xi) \frac{d^n}{dt^n} \phi(x, z; \lambda) dt \quad (13)$$

Using the expression (11) and (12) and integrating by parts to the integral on R.H.S., we have

$$D_1 \phi = \frac{d^n \phi}{d\xi^n} + \lambda \left[b_0(\xi) \frac{d^{n-1}\phi}{d\xi^{n-1}} + b_1(\xi) \frac{d^{n-2}\phi}{d\xi^{n-2}} + \dots + b_{n-1}(\xi) \phi \right] = 0 \quad (14)$$

Thus the auxiliary function $\phi(x, \xi; \lambda)$ is the integral of the linear equation $D_1 \phi = 0$ which satisfies the Cauchy conditions. Hence the resolvent kernel is of the form

$$R(x, \xi; \lambda) = -\frac{1}{\lambda} \frac{d^n}{d\xi^n} \phi(\xi, x; \lambda) \quad (15)$$

Example: Find the resolvent kernels of integral equations with the following kernel; ($\lambda = 1$)

$$K(x, \xi) = -2 + 3(x - \xi); \lambda = 1$$

Solution: Here $K(x, \xi) = -2 + 3(x - \xi); \lambda = 1$

Comparing with the relation, we have

$$a_0(x) = -2, \quad a_1(x) = 3; \text{ and all the other } a_v(x) = 0$$

The differential equation (8) reduces to

$$\frac{d^2 \phi}{dx^2} + 2 \frac{d\phi}{dx} - 3\phi = 0 \quad (1)$$

with the conditions

$$\phi = 0 \text{ at } x = \xi, \quad \frac{d\phi}{dx} = 1 \text{ at } x = \xi \quad (2)$$

The solution of equation (1) is given by

$$\phi = A(\xi)e^{-3x} + B(\xi)e^x \quad (3)$$

From (2) and (3), we obtain

$$\phi(x, \xi; 1) = \frac{1}{4}e^{x-\xi} - \frac{9}{4}e^{-3(x-\xi)}$$

Hence the resolvent kernel is given by

$$R(x, \xi; 1) = \frac{1}{\lambda} \frac{d^2}{dx^2} \phi(x, \xi; 1) = \frac{1}{4}e^{x-\xi} - \frac{1}{4}e^{-3(x-\xi)}$$

Example: Solve the Volterra's integral equation

$$\phi(x) = (1 - 2x - 4x^2) + \int_0^x [3 + 6(x - \xi) - 4(x - \xi)^2] \phi(\xi) d\xi$$

Solution: Here $f(x) = 1 - 2x - 4x^2$; $\lambda = 1$;

$$K(x, \xi) = 3 + 6(x - \xi) - 4(x - \xi)^2$$

Comparing with the relation (1), we have

$$a_0(x) = 3, a_1(x) = 6, a_2(x) = -8$$

The differential

$$D\phi = \frac{d^n \phi}{dx^n} - \lambda \left[a_0(x) \frac{d^{n-1} \phi}{dx^{n-1}} + a_1(x) \frac{d^{n-2} \phi}{dx^{n-2}} + \dots + a_{n-1}(x) \phi \right] = 0 \text{ reduces to}$$

$$\frac{d^3 \phi}{dx^3} - 3 \frac{d^2 \phi}{dx^2} - 6 \frac{d\phi}{dx} + 8\phi = 0 \quad (1)$$

with the conditions

$$\phi = 0 = \frac{d\phi}{dx} \text{ at } x = \xi \text{ and } \frac{d^2 \phi}{dx^2} = 1 \text{ at } x = \xi \quad (2)$$

The solution of equation (1) is given by

$$\phi(x, \xi; 1) = A(\xi)e^x + B(\xi)e^{-2x} + C(\xi)e^{4x} \quad (3)$$

From (2) and (3), we obtain

$$R(x, \xi; 1) = \frac{1}{\lambda} \frac{d^3}{dx^3} \phi(x, \xi; 1)$$

Therefore, the resolvent becomes

$$R(x, \xi; 1) = \frac{1}{\lambda} \frac{d^3}{dx^3} \phi(x, \xi; 1)$$

$$= -\frac{1}{9} [e^{x-\xi} + 4e^{-2(x-\xi)} - 32e^{4(x-\xi)}] \text{ Thus, the solution of the integral equation is given by}$$

$$\phi(x) = 1 - 2x - 4x^2 - \frac{1}{9} [e^{x-\xi} (9 + 10\xi + 4\xi^2) + e^{-2(x-\xi)} (4\xi - 8\xi^2) + e^{4(x-\xi)} (-32\xi - 32\xi^2)]_{\xi=0}^{x=x}$$

$$\text{or } \phi(x) = 1 - 2x - 4x^2 - 1 + 2x + 4x^2 + e^x = e^x$$

Example: Solve the Volterra integral equation of second kind, by using the method of successive approximation.

$$\phi(x) = (1 + x) - \int_0^x \phi(\xi) d\xi, \text{ with } \phi_0(x) = 1$$

Solution: The integral equation is given as $\phi(x) = (1+x) - \int_0^x \phi(\xi) d\xi$,

Here $f(x) = 1+x$, $K(x, \xi) = 1$ and $\lambda = -1$

The v th order approximation is given by

$$\phi_v(x) = f(x) + \lambda \int_0^x K(x, \xi) \phi_{v-1}(\xi) d\xi$$

$$\text{or } \phi_v(x) = (1+x) - \int_0^x \phi_{v-1}(\xi) d\xi,$$

Substituting $v = 1, 2, 3, \dots$ We have

$$\phi_1(x) = (1+x) - \int_0^x \phi_0(\xi) d\xi = (1+x) - \int_0^x d\xi = 1$$

$$\phi_2(x) = (1+x) - \int_0^x \phi_1(\xi) d\xi = (1+x) - x = 1$$

$$\phi_3(x) = (1+x) - \int_0^x \phi_2(\xi) d\xi = (1+x) - x = 1$$

$$\phi_v(x) = (1+x) - \int_0^x \phi_{v-1}(\xi) d\xi = 1$$

Hence the solution of the integral equation is given by

$$\phi(x) = \lim_{v \rightarrow \infty} \phi_v(x) = 1$$

2.4 Solution of the Fredholm integral equation by the method of successive substitutions.

Consider the Fredholm integral equation of second kind as

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi \quad (1)$$

where

- (i) The kernel $K(x, \xi) \neq 0$ is real and continuous in the rectangle $R: a \leq \xi \leq b$. Consider $K(x, \xi) \leq P$, where P is the maximum value in R .
- (ii) The function $F(x) \neq 0$ is real and continuous in an interval $I: a \leq x \leq b$. Consider $|F(x)| \leq Q$, when Q the maximum value is in the interval.
- (iii) λ is a non-zero numerical parameter.

Since there exists a continuous solution $\phi(x)$ so substituting the unknown function under an integral sign from the equation (1) itself, we obtain.

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \left\{ F(\xi) + \lambda \int_a^b K(\xi, \xi_1) \phi(\xi_1) d\xi_1 \right\} d\xi$$

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) \phi(\xi_1) d\xi_1 d\xi$$

Proceeding in this manner successively for $\phi(\xi)$, we get

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$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) F(\xi) d\xi +$$

$$\lambda^2 \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) \left\{ F(\xi_1) + \lambda \int_a^b K(\xi_1, \xi_2) \phi(\xi_2) d\xi_2 \right\} d\xi_1 d\xi$$

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) F(\xi_1) d\xi_1 d\xi +$$

$$\lambda^3 \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) \int_a^b K(\xi_1, \xi_2) \phi(\xi_2) d\xi_2 d\xi_1 d\xi + \dots +$$

$$\lambda^n \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) \int_a^b K(\xi_1, \xi_2) \dots \int_a^b K(\xi_{n-2}, \xi_{n-1}) F(\xi_{n-1}) d\xi_{n-1} \dots d\xi_1 d\xi$$

$$+ \lambda^{n+1} \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) \dots \int_a^b K(\xi_{n-1}, \xi_n) F(\xi_n) d\xi_n \dots d\xi_1 d\xi \quad (2)$$

Consider the infinite series

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) F(\xi_1) d\xi_1 d\xi + \dots \quad (3)$$

As the kernel $K(x, \xi)$ and the known function $F(\xi)$ are real and continuous, so each term of the

Above series represents a continuous function in I , provided it converges uniformly in that interval.

Since $|K(x, \xi)| \leq P$ and $|F(x)| \leq Q$

Contains the maximum value in R and I respectively.

Assume $S_n(x) = \lambda^n \int_a^b K(x, \xi)$

$$\int_a^b K(\xi, \xi_1) \dots \int_a^b K(\xi_{n-2}, \xi_{n-1}) F(\xi_{n-1}) d\xi_{n-1} \dots d\xi_1 d\xi$$

Then $|S_n(x)| \leq |\lambda^n| Q P^n (b-a)^n$

It will converge only if

$$|\lambda| P (b-a) < 1 \Rightarrow |\lambda| < \frac{1}{P(b-a)}$$

Thus the series (2) converges absolutely and uniformly when the relation (3) holds

Again, let $S_{n+1}(x) = \lambda^{n+1} \int_a^b K(x, \xi)$

$$\int_a^b K(\xi, \xi_1) \dots \int_a^b K(\xi_{n-1}, \xi_n) F(\xi_n) d\xi_n \dots d\xi_1 d\xi$$

or $|S_{n+1}(x)| < |\lambda^{n+1}| M P^{n+1} (b-a)^{n+1}$

Where M is the maximum value of the absolute value of the function $\phi(x)$ in I .

If $|\lambda|P(b-a) < 1$ then $\lim_{n \rightarrow \infty} S_{n+1}(x) = 0$

Thus we notice that the function $\phi(x)$ which satisfies the relation (2) is the continuous function

given by the series (1). We can verify the direct substitution that the function $\phi(x)$ defined by (2)

satisfies the integral equation (1). Multiplying (2) both the sides with $\lambda K(x, \xi)$ and integrating

term by term within the fixed domain, we have

$$\lambda \int_a^b K(x, \xi) \phi(\xi) d\xi = \lambda \int_a^b K(x, \xi) [F(\xi) + \lambda \int_a^b K(\xi, \xi_1) F(\xi_1) d\xi_1 + \dots + \dots] d\xi$$

The R.H.S. may be expressed as

$$= \lambda \int_a^b K(x, \xi) F(\xi) d\xi + \lambda^2 \int_a^b K(x, \xi) \int_a^b K(\xi, \xi_1) F(\xi_1) d\xi_1 d\xi + \dots$$

$$= \phi(x) - F(x)$$

$$\Rightarrow \phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi$$

2.5 Iterated Kernels

The n th iterated kernel $K_n(x, \xi)$ is determined by the relation

$$K_n(x, \xi) = \int_a^b K_m(x, z) K_{n-m}(z, \xi) dz \quad m < n \quad (1)$$

We know that $K_1(x, \xi) = K(x, \xi)$

$$\text{And } K_n(x, \xi) = \int_a^b K(x, z) K_{n-1}(z, \xi) dz \quad n = 2, 3, 4, \dots \quad (2)$$

$$\text{or } K_n(x, \xi) = \int_a^b K(x, z_1) K_{n-1}(z_1, \xi) dz_1 \quad (3)$$

Substituting $(n-1)$ for n in the equation (2), we have

$$K_{n-1}(x, \xi) = \int_a^b K(x, z) K_{n-2}(z, \xi) dz$$

$$\text{or } K_{n-1}(x, \xi) = \int_a^b K(z_1, z_2) K_{n-2}(z_2, \xi) dz_2 \quad (4)$$

Substituting $K_{n-1}(z_1, \xi)$ in the equation (3), we have

$$K_n(x, \xi) = \int_a^b K(x, z_1) \int_a^b K(z_1, z_2) K_{n-2}(z_2, \xi) dz_2 dz_1$$

$$\text{or } K_n(x, \xi) = \int_a^b \int_a^b K(x, z_1) K(z_1, z_2) K_{n-2}(z_2, \xi) dz_2 dz_1 \quad (5)$$

Substituting $(n-1)$ for n in the equation (4), we have

Put Your Own Notes

$$K_{n-2}(z_2, \xi) = \int_a^b K(z_2, z_3) K_{n-3}(z_3, \xi) dz_3 \quad (6)$$

From the relation (5) and (6), we have

$$K_n(x, \xi) = \int_a^b \int_a^b \int_a^b K(x, z_1) K(z_1, z_2) K(z_2, z_3) K_{n-3}(z_3, \xi) dz_3 dz_2 dz_1 \quad (7)$$

Performing this operation successively, we have

$$K_n(x, \xi) = \int_a^b \int_a^b \int_a^b \dots \int_a^b K(x, z_1) K(z_1, z_2) K(z_2, z_3) K(z_{n-1}, \xi) dz_{n-1} \dots dz_3 dz_2 dz_1$$

$$\text{or } K_n(x, \xi) = \int_a^b \int_a^b \int_a^b \dots \int_a^b K(x, z_1) K(z_1, z_2) K(z_2, z_3) \dots K(z_{m-1}, z_m) K(z_m, z_{m+1}) \dots K(z_{n-1}, \xi) dz_{n-1} \dots dz_3 dz_2 dz_1 \quad (8)$$

Similarly, we obtain

$$K_m(x, z) = \underbrace{\int_a^b \int_a^b \int_a^b \dots \int_a^b K(x, p_1) K(p_1, p_2) K(p_2, p_3) \dots K(p_{m-1}, z) dp_{m-1} \dots dp_3 dp_2 dp_1}_{(m-1)\text{th order integral}} \quad (9)$$

$$\text{and } K_{n-m}(z, \xi) = \underbrace{\int_a^b \int_a^b \int_a^b \dots \int_a^b K(z, q_1) K(q_1, q_2) K(q_2, q_3) \dots K(q_{n-m-1}, \xi) dq_{n-m-1} \dots dq_3 dq_2 dq_1}_{(n-m-1)\text{th order integral}} \quad (10)$$

From the relations (9) and (10), we have

$$\begin{aligned} & \int_a^b K_m(x, z) K_{n-m}(z, \xi) dz \\ &= \int_a^b \left\{ \underbrace{\int_a^b \int_a^b \int_a^b \dots \int_a^b K(x, p_1) K(p_1, p_2) K(p_2, p_3) K(p_{m-1}, z) dp_{m-1} \dots dp_3 dp_2 dp_1}_{(m-1)\text{th order integral}} \right\} \\ & \times \left\{ \underbrace{\int_a^b \int_a^b \dots \int_a^b K(z, q_1) K(q_1, q_2) \dots K(q_{n-m-1}, \xi) dq_{n-m-1} \dots dq_3 dq_2 dq_1}_{(n-m-1)\text{th order integral}} \right\} dz \end{aligned}$$

$$\begin{aligned} & \text{Or } \int_a^b K_m(x, z) K_{n-m}(z, \xi) dz \\ &= \underbrace{\int_a^b \int_a^b \int_a^b \dots \int_a^b K(x, p_1) K(p_1, p_2) K(p_{m-1}, z) K(z, q_1) K(q_1, q_2) \dots K(q_{n-m-1}, \xi) dq_{n-m-1} \dots dq_3 dq_2 dq_1 dp_{m-1} \dots dp_3 dp_2 dp_1}_{(n-1)\text{th order integral}} \quad (11) \end{aligned}$$

Now without changing the limits of integration, the variables of integration are changed as follows:

$$p_1 \ p_2 \ p_3 \ \dots \ p_{m-1} \ z \ q_1 \ q_2 \ q_3 \ \dots \ q_{n-m-1}$$

$$z_1 \ z_2 \ z_3 \dots z_{m-1} \ z_m \ z_{m+1} \ z_{m+2} \ z_{m+3} \dots z_{n-1}$$

Thus we have

$$\int_a^b K_m(x, z) K_{n-m}(z, \xi) dz =$$

$$\underbrace{\int_a^b \int_a^b \dots \int_a^b K(x, z_1) K(z_1, z_2) K(z_2, z_3) \dots K(z_{m-1}, z_m)}_{(n-1)\text{th order integral}}$$

$$K(z_m, z_{m+1}) \dots K(z_{n-1}, \xi) dz_{n-1} dz_m \dots dz_2 dz_1 \quad (12)$$

Hence from the relations (9) and (12), we have

$$K_n(x, z) = \int_a^b K_m(x, z) K_{n-m}(z, \xi) dz, \quad m < n$$

2.6 Solution of Fredholm's Equation

A function $K(x, \xi)$ reciprocal to $K(x, \xi)$ exists where $K(x, \xi)$ is real and continuous in R , $K(x, \xi) \neq 0$, $f(x)$ is real and continuous in I , $f(x) \neq 0$, then the Fredholm's integral equation of second kind has one and only one continuous solution in the interval I , $a \leq x \leq b$.

Consider the Fredholm integral equation of second kind as

$$\phi(x) = f(x) + \int_a^b K(x, \xi) \phi(\xi) d\xi \quad (1)$$

Where the reciprocal function $K(x, \xi)$ of $K(x, \xi)$ is known. Let the equation (1) has a continuous

solution $\phi(x)$, then

$$f(\xi) = f(\xi) + \int_a^b K(\xi, \xi_1) \phi(\xi_1) d\xi_1$$

Multiplying by $K(x, \xi)$ and integrating over the interval (a, b) , we have

$$\int_a^b k(x, \xi) \phi(\xi) d\xi = \int_a^b k(x, \xi) f(\xi) d\xi + \int_a^b \int_a^b k(x, \xi) K(\xi, \xi_1) \phi(\xi_1) d\xi_1 d\xi$$

$$\Rightarrow \int_a^b k(x, \xi) \phi(\xi) d\xi = \int_a^b k(x, \xi) f(\xi) d\xi + \int_a^b [K(x, \xi_1) + k(x, \xi_1)] \phi(\xi_1) d\xi_1$$

$$\Rightarrow 0 = \int_a^b k(x, \xi) f(\xi) d\xi + \int_a^b K(x, \xi_1) \phi(\xi_1) d\xi_1$$

$$\text{Again } \int_a^b k(x, \xi) \phi(\xi_1) d\xi_1 = -(\phi(x) - f(x));$$

$$\text{since } -\int_a^b K(x, \xi) \cdot \phi(\xi) d\xi = -f(x) \phi(x)$$

$$\Rightarrow \phi(x) = -\left(f(x) + \int_a^b K(x, \xi) f(\xi) d\xi\right) \quad (2)$$

It follows that the integral equation (1) has a continuous solution and is given by the equation (2) which is unique.

Example: Find the iterated kernels for the following kernels.

$$K(x, \xi) = x - \xi, \text{ if } a=0, b=1$$

Solution: Using formulas of integral kernels, we obtain in succession:

$$K_1(x, \xi) = K(x, \xi) = x - \xi$$

$$\text{or } K_2(x, \xi) = \int_0^1 (x-z)(z-\xi) dz = \frac{x+\xi}{2} - x\xi - \frac{1}{3}$$

$$\text{or } K_3(x, \xi) = \int_0^1 (x-z) \left(\frac{z+\xi}{2} - z\xi - \frac{1}{3} \right) dz = (-1) \left(\frac{x-\xi}{12} \right)$$

$$\text{or } K_4(x, \xi) = -\frac{1}{12} \int_0^1 (x-z)(z-\xi) dz = (-1)^2 \frac{1}{12} \left(\frac{x+\xi}{2} - x\xi - \frac{1}{3} \right)$$

$$\text{or } K_4(x, \xi) = -\int_0^1 (x-z) \left(\frac{z+\xi}{2} - z\xi - \frac{1}{3} \right) dz = (-1) \left(\frac{x-\xi}{12^2} \right)$$

$$\text{or } K_6(x, \xi) = \frac{1}{12^2} \int_0^1 (x-z)(z-\xi) dz = (-1)^2 \frac{1}{12^2} \left(\frac{x+\xi}{2} - x\xi - \frac{1}{3} \right)$$

and so on

It follows that the iterated kernels are of the form

$$(i) \text{ If } n=2v-1 \text{ then } K_{2v-1}(x, \xi) = \frac{(-1)^v}{12^{v-1}} (x-\xi)$$

$$(ii) \text{ If } n=2v \text{ then } K_{2v}(x, \xi) = \frac{(-1)^{v-1}}{12^{v-1}} \left(\frac{x+\xi}{2} - x\xi - \frac{1}{3} \right), v=1, 2, 3, \dots$$

Example: Solve the following linear integral equation

$$\phi(x) = x + \int_0^{1/2} \phi(\xi) d\xi$$

Solution: Here $K(x, \xi) = 1$, $\lambda = 1$ and $F(x) = x$

We know that $K_1(x, \xi) = K(x, \xi) = 1$

$$\text{or } K_2(x, \xi) = \int_0^{1/2} \xi dz = \frac{1}{2}$$

$$\text{or } K_3(x, \xi) = \int_0^{1/2} dz = \left(\frac{1}{2} \right)^2$$

$$\text{or } K_n(x, \xi) = \left(\frac{1}{2} \right)^{n-1}$$

Thus the resolving kernel is given by

$$R(x, \xi; 1) = \sum_{n=1}^{\infty} \left(\frac{\lambda}{2} \right)^{n-1} = \frac{2}{2-\lambda}$$

$$\text{or } R(x, \xi; 1) = 2, \text{ as } \lambda = 1$$

Hence the solution of the integral equation is given by

$$\phi(x) = F(x) + \lambda \int_a^b R(x, \xi; \lambda) F(\xi) d\xi$$

$$\text{or } \phi(x) = x + \int_0^{1/2} 2\xi d\xi = x + \frac{1}{4} = x + \text{constant}$$

Example: Solve the integral equations

$$\phi(x) = f(x) + \lambda \int_0^1 e^{x-\xi} \phi(\xi) d\xi$$

$$\phi(x) = 1 + \lambda \int_0^\pi \sin(x+\xi) \phi(\xi) d\xi$$

Solution : (i) Here $K_1(x, \xi) = K(x, \xi) = e^{x-\xi}$

$$\text{or } K_2(x, \xi) = \int_0^1 e^{x-t} e^{t-\xi} dt = e^{x-\xi}$$

Proceeding in this way, we find that all the iterated kernels coincide with $K(x, \xi)$. Thus, the

resolvent kernel becomes

$$\Gamma(x, \xi; \lambda) = K(x, \xi)(1 + \lambda + \lambda^2 + \dots) = \frac{e^{(x-\xi)}}{(1-\lambda)}$$

The resolvent kernel is an analytic function of λ except at the point $\lambda = 1$, which is a simple pole

of the kernel Γ .

The solution of the equation is given by

$$\phi(x) = f(x) - \frac{\lambda}{\lambda-1} \int_0^1 e^{x-\xi} f(\xi) d\xi$$

(ii) Here $K_1(x, \xi) = K(x, \xi) = \sin(x+\xi)$

$$\text{or } K_2(x, \xi) = \int_0^\pi \sin(x+t) \sin(t+\xi) dt = \frac{1}{2} \pi \cos(x-\xi)$$

$$\text{or } K_3(x, \xi) = \frac{\pi}{2} \int_0^\pi \sin(x+t) \cos(t-\xi) dt = \left(\frac{\pi}{2}\right)^2 \sin(x+\xi)$$

Proceeding in this way, we have

$$K_4(x, \xi) = \left(\frac{\pi}{2}\right)^3 \cos(x-\xi), \quad K_5(x, \xi) = \left(\frac{\pi}{2}\right)^4 \sin(x+\xi)$$

$$K_6(x, \xi) = \left(\frac{\pi}{2}\right)^5 \cos(x-\xi), \quad K_7(x, \xi) = \left(\frac{\pi}{2}\right)^6 \sin(x+\xi).$$

The resolvent kernel $R(x, \xi; \lambda)$ is given by

$$R(x, \xi; \lambda) = \sum_{n=1}^{\infty} \lambda^{n-1} K_n(x, \xi)$$

or

$$R(x, \xi; \lambda) = K_1(x, \xi) + \lambda K_2(x, \xi) + \lambda^2 K_3(x, \xi) + \lambda^3 K_4(x, \xi) + \lambda^4 K_5(x, \xi) + \lambda^5 K_6(x, \xi) + \dots$$

$$\text{or } R(x, \xi; \lambda) = \sin(x + \xi) \left\{ 1 + \left(\frac{\lambda\pi}{2} \right)^2 + \left(\frac{\lambda\pi}{2} \right)^4 + \dots \right\} + \frac{\lambda\pi}{2} \cos(x - \xi) \left\{ 1 + \left(\frac{\lambda\pi}{2} \right)^2 + \left(\frac{\lambda\pi}{2} \right)^4 + \dots \right\}$$

$$\text{or } R(x, \xi; \lambda) = \left[\sin(x + \xi) + \frac{\lambda\pi}{2} \cos(x - \xi) \right] \left\{ 1 + \left(\frac{\lambda\pi}{2} \right)^2 + \left(\frac{\lambda\pi}{2} \right)^4 + \dots \right\}$$

$$R(x, \xi; \lambda) = \left[\sin(x + \xi) + \frac{\lambda\pi}{2} \cos(x - \xi) \right] \frac{1}{1 - \left(\frac{\lambda\pi}{2} \right)^2}$$

$$\text{Provided } \left| \frac{\lambda\pi}{2} \right| < 1 \Rightarrow |\lambda| < \frac{2}{\pi}$$

$$\text{or } R(x, \xi; \lambda) = \frac{2}{4 - \lambda^2 \pi^2} [2 \sin(x + \xi) + \lambda\pi \cos(x - \xi)]$$

Thus the solution of the integral equation is given by

$$\phi(x) = f(x) + \lambda \int_0^\pi R(x, \xi; \lambda) f(\xi) d\xi$$

$$\Rightarrow \phi(x) = 1 + \frac{2\lambda}{4 - \lambda^2 \pi^2} \int_0^\pi [2 \sin(x + \xi) + \lambda\pi \cos(x - \xi)] d\xi$$

$$\Rightarrow \phi(x) = 1 + \frac{2\lambda}{4 - \lambda^2 \pi^2} [-2 \cos(x + \xi) - \lambda\pi \sin(x - \xi)]_0^\pi$$

$$\Rightarrow \phi(x) = 1 + \frac{2\lambda}{4 - \lambda^2 \pi^2} [2 \cos x + \lambda\pi \sin x]; \text{ where } |\lambda| < \frac{2}{\pi}$$

Example: Solve the non-homogeneous Fredholm integral equation of the second kind, by the method of successive approximations to the third order.

$$\phi(x) = 2x + \lambda \int_0^1 (x + \xi) \phi(\xi) d\xi, \quad \phi_0(x) = 1$$

Solution: The integral equation is given as

$$\phi(x) = 2x + \lambda \int_0^1 (x + \xi) \phi(\xi) d\xi, \quad \phi_0(x) = 1 \quad (1)$$

Consider $\phi_n(x)$ denotes the n th order approximation then, we have

$$\phi_n(x) = 2x + \lambda \int_0^1 (x + \xi) \phi_{n-1}(\xi) d\xi \quad (2)$$

Substituting $n=1, 2, 3, 4, \dots$ in the equation (2), we have

$$\phi_1(x) = 2x + \lambda \int_0^1 (x + \xi) \phi_0(\xi) d\xi = 2x + \lambda \int_0^1 (x + \xi) d\xi$$

$$\text{or } \phi_1(x) = 2x + \lambda \left(x + \frac{1}{2} \right)$$

$$\text{Also } \phi_2(x) = 2x + \lambda \int_0^1 (x + \xi) \phi_1(\xi) d\xi$$

$$\text{or } \phi_2(x) = 2x + \lambda \int_0^1 (x + \xi) \left\{ 2\xi + \lambda \left(\xi + \frac{1}{2} \right) \right\} d\xi$$

$$\text{or } \phi_2(x) = 2x + \lambda \int_0^1 (x + \xi) \left\{ \frac{1}{2} \lambda + \xi(2 + \lambda) \right\} d\xi$$

$$\text{or } \phi_2(x) = 2x + \lambda \left(x + \frac{2}{3} \right) + \lambda^2 \left(x + \frac{7}{12} \right). \quad (3)$$

$$\text{and } \phi_3(x) = 2x + \lambda \int_0^1 (x + \xi) \phi_2(\xi) d\xi$$

$$\text{or } \phi_3(x) = 2x + \lambda \int_0^1 (x + \xi) \left\{ 2\xi + \lambda \left(\xi + \frac{2}{3} \right) + \lambda^2 \left(\xi + \frac{7}{12} \right) \right\} d\xi$$

$$\text{or } \phi_3(x) = 2x + \lambda \int_0^1 (x + \xi) \left\{ \frac{2\lambda}{3} + \frac{7\lambda^2}{12} + (2 + \lambda + \lambda^2)\xi \right\} d\xi$$

$$\text{or } \phi_3(x) = 2x + \lambda$$

$$\int_0^1 \left[x \left(\frac{2\lambda}{3} + \frac{7\lambda^2}{12} \right) + \xi \left(\frac{2\lambda}{3} + \frac{7\lambda^2}{12} + 2x + \lambda x + \lambda^2 x + \xi(2 + \lambda + \lambda^2) \right) \right] d\xi$$

$$\text{or } \phi_3(x) = 2x + \lambda \left(x + \frac{2}{3} \right) + \lambda^2 \left(\frac{7}{6}x + \frac{2}{3} \right) + \lambda^3 \left(\frac{13}{12}x + \frac{5}{8} \right)$$

EXERCISE-2

- Find the resolvent kernel's of the Volterra's integral equations with the following kernels.

$$(a.) K(x, \xi) = \begin{cases} e^{x-\xi} & x > \xi \\ 0 & \xi > x \end{cases}$$

$$(b.) K(x, \xi) = \frac{2 + \cos x}{2 + \cos \xi}$$

$$(c.) K(x, \xi) = 2 - (x - \xi); \lambda = 1$$

$$(d.) K(x, \xi) = 2x; \lambda = 1$$

- Solve the Volterra's integral equations by finding resolvent kernel

$$(a.) \phi(x) = (\cos x - x - 2) + \int_0^x (\xi - x) \phi(\xi) d\xi$$

$$(b.) \phi(x) = 29 + 6x + \int_0^x [5 - 6(x - \xi)] \phi(\xi) d\xi$$

$$(c.) \phi(x) = 1 + x^2 + \int_0^x \frac{1 + x^2}{1 + \xi^2} \phi(\xi) d\xi$$

$$(d.) \phi(x) = 1 + \int_0^x x\xi \phi(\xi) d\xi$$

3. Solve the Volterra integral equations of second kind, by using the method of successive approximation.

(a.) $\phi(x) = x - \int_0^x (x-\xi) \phi(\xi) d\xi$ with $\phi_0(x) = 0$

(b.) $\phi(x) = 1 + \int_0^x (x-\xi) \phi(\xi) d\xi$ with $\phi_0(x) = 0$

(c.) $\phi(x) = 1 + \int_0^x \phi(\xi) d\xi$ with $\phi_0(x) = 0$

(d.) $\phi(x) = 1 + x + \int_0^x (x-\xi) \phi(\xi) d\xi$ with $\phi_0(x) = 0$

4. Find the iterated kernels for the following kernels.

(a.) $K(x, \xi) = x + \sin \xi$; $a = -\pi$, $b = \pi$

(b.) $K(x, \xi) = e^x \cos \xi$; $a = 0$, $b = \pi$

5. Solve the following integral equations

(a.) $\phi(x) = \left(\frac{3}{2}e^x - \frac{1}{2}xe^x - \frac{1}{2} \right) + \frac{1}{2} \int_0^1 \xi \phi(\xi) d\xi$

(b.) $\phi(x) = \frac{5x}{6} + \frac{1}{2} \int_0^1 x\xi \phi(\xi) d\xi$

(c.) $\phi(x) = \left(e^x - \frac{1}{2}e + \frac{1}{2} \right) + \frac{1}{2} \int_0^1 \phi(\xi) d\xi$

(d.) $\phi(x) = 1 + \lambda \int_0^1 (x+\xi) \phi(\xi) d(\xi)$, $\phi_0(x) = 1$

Find the solution by using successive approximation to the third order.

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CHAPTER 3

FREDHOLM INTEGRAL EQUATION

3.1 Fredholm Theorems

Fredholm First Theorem

The non-homogeneous Fredholm integral equation of second kind.

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi,$$

Under the assumption that the function $F(x)$ and $K(x, \xi)$ are integrable has a unique solution, is of the form

$$\phi(x) = F(x) + \lambda \int_a^b R(x, \xi; \lambda) F(\xi) d\xi,$$

Where the resolvent kernel R is a meromorphic function of the parameter, being the ratio of two entire function of the parameter λ

$$R(x, \xi; \lambda) = D(x, \xi; \lambda) / D(\lambda), D(\lambda) \neq 0$$

Defined by Fredholm's series of the form

$$D(\lambda) = 1 + \sum_{m=1}^{\infty} \frac{(-\lambda)^m}{m!} \int_a^b \dots \int_a^b K \left(\begin{matrix} \xi_1 & \xi_2 & \dots & \xi_m \\ \xi_1 & \xi_2 & \dots & \xi_m \end{matrix} \right) d\xi_1 d\xi_2 \dots d\xi_m$$

And

$$D(x, \xi; \lambda) = K(x, \xi) + \sum_{m=1}^{\infty} \frac{(-\lambda)^m}{m!} \int_a^b \dots \int_a^b K \left(\begin{matrix} \xi_1 & \xi_2 & \dots & \xi_m \\ \xi_1 & \xi_2 & \dots & \xi_m \end{matrix} \right) d\xi_1 d\xi_2 \dots d\xi_m$$

These series converge for all values of λ . In particular, the solution of the homogeneous integral equation is zero. Where

$$K \left(\begin{matrix} \xi_1 & \xi_2 & \dots & \xi_m \\ \xi_1 & \xi_2 & \dots & \xi_m \end{matrix} \right) = \begin{vmatrix} K(\xi_1, \xi_1) & K(\xi_1, \xi_2) & \dots & K(\xi_1, \xi_m) \\ K(\xi_2, \xi_1) & K(\xi_2, \xi_2) & \dots & K(\xi_2, \xi_m) \\ \vdots & \vdots & \ddots & \vdots \\ K(\xi_m, \xi_1) & K(\xi_m, \xi_2) & \dots & K(\xi_m, \xi_m) \end{vmatrix}$$

and

$$K \left(\begin{matrix} x & \xi_1 & \xi_2 & \dots & \xi_m \\ \xi & \xi_1 & \xi_2 & \dots & \xi_m \end{matrix} \right) = \begin{vmatrix} K(x, \xi) & K(x, \xi_1) & K(x, \xi_2) & \dots & K(x, \xi_m) \\ K(\xi_1, \xi) & K(\xi_1, \xi_1) & K(\xi_1, \xi_2) & \dots & K(\xi_1, \xi_m) \\ K(\xi_2, \xi) & K(\xi_2, \xi_1) & K(\xi_2, \xi_2) & \dots & K(\xi_2, \xi_m) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ K(\xi_m, \xi) & K(\xi_m, \xi_1) & K(\xi_m, \xi_2) & \dots & K(\xi_m, \xi_m) \end{vmatrix}$$

Note:

- (a) Every zero of Fredholm function $D(\lambda)$ is a pole of the resolvent kernel

$$R(\xi, \xi; \lambda) = D(\xi, \xi; \lambda) / D(\lambda)$$

The order of this pole is at most equal to the order of the zero of denominator $D(\lambda)$.

- (b) The zero's of the function $D(\lambda)$ are called the eigen values of the kernel $K(x, \xi)$, since

$D(0) = 1$, therefore zero is never an eigen value. The set of all eigen values of this kernel is known the spectrum of the integral equation.

Fredholm Second Theorem

If λ_0 is a zero of multiplicity m of the function $D(\lambda)$, then the Homogeneous integral equation

$$\phi(x) = \lambda_0 \int_a^b K(x, \xi) \phi(\xi) d\xi,$$

possesses atleast one, and at most m , linearly independent solutions

$$\phi_i(x) = D_v \begin{pmatrix} x_1, x_2, \dots, x_{i-1}, x, x_{i+1}, \dots, x_v \\ \xi_1, \xi_2, \dots, \xi_v \end{pmatrix}; \lambda_0 \quad (i=1, 2, \dots, v; 1 \leq v \leq m)$$

not identically zero, and any other solution of this equation is a linear combination of these solutions.

Fredholm's Third Theorem

For the non-homogeneous integral equation of second kind

$$\phi(x) = f(x) + \lambda_0 \int_a^b K(x, \xi) \phi(\xi) d\xi,$$

to possess a solution in the case $D(\lambda_0) = 0$, it is necessary and sufficient that the given function $f(x)$ be orthogonal to all the eigen solutions, $\psi_i(x)$, $i=1, 2, \dots, v$ of the associated homogeneous equation corresponding to the eigen value λ_0 and forming the fundamental system.

Example: Using Fredholm determinants, find the resolvent kernel of the following kernel.

$$K(x, \xi) = xe^\xi; a=0, b=1$$

Solution: Here $B_0(x, \xi) = K(x, \xi) = xe^\xi$

$$B_1(x, \xi) = \int_0^1 \begin{vmatrix} K(x, \xi) & K(x, \xi_1) \\ K(\xi_1, \xi) & K(\xi_1, \xi_1) \end{vmatrix} d\xi_1 = \int_0^1 \begin{vmatrix} xe^\xi & xe^{\xi_1} \\ \xi_1 e^\xi & \xi_1 e^{\xi_1} \end{vmatrix} d\xi_1 = 0$$

$$\text{and } B_2(x, \xi) = \int_0^1 \int_0^1 \begin{vmatrix} K(x, \xi) & K(x, \xi_1) & K(x, \xi_2) \\ K(\xi_1, \xi) & K(\xi_1, \xi_1) & K(\xi_1, \xi_2) \\ K(\xi_2, \xi) & K(\xi_2, \xi_1) & K(\xi_2, \xi_2) \end{vmatrix} d\xi_1 d\xi_2$$

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$$\text{or } B_2(x, \xi) = \int_0^1 \int_0^1 \begin{vmatrix} xe^{\xi} & xe^{\xi_1} & xe^{\xi_2} \\ \xi_1 e^{\xi} & \xi_1 e^{\xi_1} & \xi_1 e^{\xi_2} \\ \xi_2 e^{\xi} & \xi_2 e^{\xi_1} & \xi_2 e^{\xi_2} \end{vmatrix} d\xi_1 d\xi_2 = 0$$

It follows that all subsequent $B_m(x, \xi) = 0$ for $m \geq 3$.

$$\text{also } C_1 = \int_0^1 K(\xi_1, \xi_1) d\xi_1 = \int_0^1 \xi_1 e^{\xi_1} d\xi_1 = 1$$

$$\text{and } C_2 = \int_0^1 \int_0^1 \begin{vmatrix} K(\xi_1, \xi_1) & K(\xi_1, \xi_2) \\ K(\xi_2, \xi_1) & K(\xi_2, \xi_2) \end{vmatrix} d\xi_1 d\xi_2$$

$$\text{or } C_2 = \int_0^1 \int_0^1 \begin{vmatrix} \xi_1 e^{\xi_1} & \xi_1 e^{\xi_2} \\ \xi_2 e^{\xi_1} & \xi_2 e^{\xi_2} \end{vmatrix} d\xi_1 d\xi_2 = 0$$

$\Rightarrow$ that all subsequent C_m also vanish

Hence, we have

$$D(x, \xi; \lambda) = K(x, \xi) + \sum_{m=1}^{\infty} \frac{(-1)^m}{m!} \lambda^m B_m(x, \xi) = xe^{\xi}$$

$$\text{and } D(\lambda) = 1 + \sum_{m=1}^{\infty} \frac{(-\lambda)^m}{m!} C_m = 1 - \lambda.$$

Thus the Fredholm resolvent kernel is given by

$$R(x, \xi; \lambda) = \frac{D(x, \xi, \lambda)}{D(\lambda)} = \frac{xe^{\xi}}{1 - \lambda}$$

Example: Using Fredholm determinants, find the resolvent kernel of the following kernel.

$$K(x, \xi) = \sin x \cos \xi; \quad 0 \leq x \leq 2\pi, \quad 0 \leq \xi \leq 2\pi$$

Solution: Here $B_0(x, \xi) = K(x, \xi) = \sin x \cos \xi$

$$B_1(x, \xi) = \int_0^{2\pi} \begin{vmatrix} \sin x \cos \xi & \sin x \cos \xi_1 \\ \sin \xi_1 \cos \xi & \sin \xi_1 \cos \xi_1 \end{vmatrix} d\xi_1 = 0, \quad B_2(x, \xi) = 0$$

It follows that all subsequent $B_2(x, \xi) = 0$

$$\text{also } C_1 = \int_0^{2\pi} \sin \xi_1 \cos \xi_1 d\xi_1 = 0, \quad C_2 = 0,$$

Which shows that all other C_m are also zero

Hence, we have

$$D(x, \xi; \lambda) = K(x, \xi) + \sum_{m=1}^{\infty} \frac{(-1)^m}{m!} \lambda^m B_m(x, \xi) = \sin x \cos \xi$$

$$\text{And } D(\lambda) = 1 + \sum_{m=1}^{\infty} \frac{(-\lambda)^m}{m!} C_m = 1$$

Thus the Fredholm resolvent kernel is given by

$$R(x, \xi; \lambda) = \frac{D(x, \xi; \lambda)}{D(\lambda)} = \sin x - \cos \xi.$$

Example: Find the resolvent kernel of the following kernel.

$$K(x, \xi) = x - 2\xi; \quad 0 \leq x \leq 1, \quad 0 \leq \xi \leq 1$$

Solution: Here $B_0(x, \xi) = C_0 K(x, \xi) = x - 2\xi$ and $C_0 = 1$

The further coefficients may be determined by using the recursive relations.

$$B_m(x, \xi) = C_m K(x, \xi) - m \int_a^b K(x, \xi_1) B_{m-1}(\xi_1, \xi) d\xi_1 \quad (1)$$

$$\text{and } C_m = B_{m-1}(\xi_1, \xi_1) d\xi_1 \quad (2)$$

For the relation (2), we have

$$C_1 = \int_a^b B_0(\xi_1, \xi_1) d\xi_1 = \int_0^1 (-\xi_1) d\xi_1 = -\frac{1}{2}$$

From the relation (1), we have

$$B_1(x, \xi) = C_1 K(x, \xi) - \int_0^1 K(x, \xi_1) B_0(\xi_1, \xi) d\xi_1$$

$$B_1(x, \xi) = -\frac{1}{2}(x - 2\xi) - \int_0^1 (x - 2\xi_1)(\xi_1 - 2\xi) d\xi_1$$

$$\text{or } B_1(x, \xi) = -x - \xi + 2x\xi + \frac{2}{3}$$

$$\text{Also } C_2 = \int_0^1 \left(-2\xi_1 + 2\xi_1^2 + \frac{2}{3} \right) d\xi_1 = \frac{1}{3}$$

$$\text{And } B_2(x, \xi) = \frac{x - 2\xi}{3} - 2 \int_0^1 (x - 2\xi_1) \left(-\xi_1 - \xi + 2\xi_1\xi + \frac{2}{3} \right) d\xi_1 = 0$$

$$\Rightarrow C_3 = C_4 = \dots = 0, \text{ and } B_3(x, \xi) = B_4(x, \xi) = \dots = 0$$

$$\text{Hence } K(x, \xi) C_m = 1 + \frac{1}{2}\lambda + \left(\frac{1}{6}\right)\lambda^2.$$

$$\text{And } D(x, \xi; \lambda) = K(x, \xi) + \sum_{m=1}^{\infty} \frac{(-\lambda)^m}{m!} B_m(x, \xi)$$

$$\text{or } D(x, \xi; \lambda) = x - 2\xi + \left(x + \xi - 2x\xi - \frac{2}{3} \right) \lambda$$

Thus the resolvent kernel is given by

$$R(x, \xi; \lambda) = \frac{D(x, \xi; \lambda)}{D(\lambda)}$$

$$\text{or } R(x, \xi; \lambda) = \frac{x - 2\xi + \lambda \left(x + \xi - 2x\xi - \frac{2}{3} \right)}{1 + \frac{1}{2}\lambda + \frac{1}{6}\lambda^2}.$$

Example: Solve the following integral equation.

$$\phi(x) = e^x + \lambda \int_0^1 2e^x e^\xi \phi(\xi) d\xi$$

Solution: $\phi(x) = e^x + \lambda \int_0^1 2e^x e^\xi \phi(\xi) d\xi$

Here $K(x, \xi) = 2e^x e^\xi$, $f(x) = e^x$

The resolvent kernel of the kernel $K(x, \xi)$ is obtained as

$$R(x, \xi; \lambda) = \frac{D(x, \xi; \lambda)}{D(\lambda)} = \frac{2e^x e^\xi}{1 - (e^2 - 1)\lambda}$$

Thus the solution of the integral equation is given by

$$\phi(x) = e^x + \frac{2\lambda e^x}{1 - (e^2 - 1)\lambda} \int_0^1 e^\xi \cdot e^\xi d\xi$$

$$\phi(x) = e^x + \frac{2\lambda e^x}{1 - (e^2 - 1)\lambda} \frac{1}{2} (e^{2\xi})_0^1$$

$$\text{or } \phi(x) = e^x + \frac{\lambda(e^2 - 1)e^x}{1 - (e^2 - 1)\lambda}$$

$$\text{or } \phi(x) = \frac{1}{1 - (e^2 - 1)\lambda} e^x$$

3.2 Fredholm's Associated Equation

Consider the non-homogeneous Fredholm integral equation of second kind.

$$\phi(x) = f(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi \quad (1)$$

By permuting the variables x and ξ in the kernel $K(x, \xi)$, we obtain a new kernel $K(\xi, x)$, in general, different from the first. The corresponding Fredholm equation.

$$\phi(x) = g(x) + \lambda \int_a^b K(\xi, x) \psi(\xi) d\xi \quad (2)$$

is said to be Fredholm associated equation with equation (1). Since $D(\lambda)$ and $D(x, \xi; \lambda)$ are the Fredholm first and second series of the equation (1) then $D(\lambda)$ and $D(x, \xi; \lambda)$ are the Fredholm series of the associated equation (2). The kernels of the equations (1) and (2) have the same eigen values as zeroes of the first Fredholm series $D(\lambda)$. Thus if

$$R(x, \xi; \lambda) = D(x, \xi; \lambda) / D(\lambda)$$

is a resolvent kernel equation (1), then

$$R(\xi, x; \lambda) = D(\xi, x; \lambda) / D(\lambda)$$

is a resolvent kernel of the associated equation (2), whose solution is given by

$$\psi(x) = g(x) + \lambda \{ D(\xi, \xi; \lambda) / D(\lambda) \} g(\xi) d\xi \quad (3)$$

Where $D(\lambda) \neq 0$

again, If $D_n \begin{pmatrix} x_1, x_2, \dots, x_n \\ \xi_1, \xi_2, \dots, \xi_n \end{pmatrix}; \lambda$

is a minor of order n of Fredholm equation (1), then

$$D_n \begin{pmatrix} \xi_1, \xi_2, \dots, \xi_n \\ x_1, x_2, \dots, x_n \end{pmatrix}; \lambda$$

is a minor of the same order of the associated equation. Thus the following equation is satisfied

$$D_v \begin{pmatrix} x_1, x_2, \dots, x_n \\ \xi_1, \xi_2, \xi_{i-1}, x, \xi_{i+1}, \dots, \xi_n \end{pmatrix}; \lambda_0 = \lambda_0 \int_a^b K(\lambda, x) D_v \begin{pmatrix} x_1, x_2, \dots, x_n \\ \xi_1, \xi_2, \xi_{i-1}, s, \xi_{i+1}, \dots, \xi_n \end{pmatrix}; \lambda_0 ds \quad (4)$$

Hence for fixed $x_1, x_2, \dots, x_n$ and $\xi_1, \xi_2, \dots, \xi_n$ the function

$$\psi_i(x) = D_v \begin{pmatrix} x_1, x_2, \dots, x_n \\ \xi_1, \dots, \xi_{i-1}, x, \xi_{i+1}, \dots, \xi_n \end{pmatrix}; \lambda_0 \quad (5)$$

is a characteristic solution of the homogeneous associated equation

$$\psi(x) = \int_a^b K(s, x) \psi(s) ds \quad (6)$$

Thus $\psi_1(x), \psi_2(x), \dots, \psi_v(x)$ are the v characteristic solutions of the associated equation

corresponding to the eigen value λ_0 , which may be written as

$$\psi_i(x) = \frac{D_v \begin{pmatrix} x_1, \dots, x_n \\ \xi_1, \dots, \xi_{i-1}, x, \xi_{i+1}, \dots, \xi_n \end{pmatrix}; \lambda_0}{D_v \begin{pmatrix} x_1, \dots, x_n \\ \xi_1, \dots, \xi_{i-1}, x, \xi_{i+1}, \dots, \xi_n \end{pmatrix}; \lambda_0} \quad (7)$$

the denominator is non-singular in nature.

Substituting x in v different places in the lower sequence of the relation (7), the fundamental

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System of v characteristic solutions.

$\psi_1(x), \psi_2(x), \dots, \psi_v(x)$ of the associated equation is obtained which are linearly independent. Every solution of the

associated homogeneous equation is a linear combination of solutions.

Example: Characteristic solutions, corresponding to distinct characteristic values of Fredholm's

integral equation and its associate equation, are orthogonal.

Since $\phi(x)$ be a characteristic solution of the homogeneous equation

$$\text{Solution: } \phi(x) = \lambda_0 \int_a^b k(x, \xi) \phi(\xi) d\xi \quad (1)$$

corresponding to a characteristic value λ_0 .

Let $\psi(x)$ be a characteristic solution of the associated equation

$$\psi(x) = \lambda_1 \int_a^b k(x, \xi) \psi(\xi) d\xi \quad (2)$$

corresponding to a characteristic value λ_1 where $\lambda_0 \neq \lambda_1$

Multiplying (1) by $\lambda_1 \psi(x)$ and (2) by $\lambda_0 \phi(x)$, integrating and then subtracting, we have

$$\begin{aligned} (\lambda_1 - \lambda_0) \int_a^b \phi(x) \psi(x) dx &= \lambda_0 \lambda_1 \int_a^b \int_a^b K(x, \xi) \psi(x) \phi(\xi) d\xi dx - \lambda_0 \lambda_1 \int_a^b \int_a^b K(\xi, x) \psi(\xi) \phi(x) d\xi dx = 0 \end{aligned}$$

upon interchanging x and ξ , the second integral is identical to the first one. Thus

$$\int_a^b \phi(x) \psi(x) dx = 0, \lambda_1 \neq \lambda_0$$

$\Rightarrow$ That the characteristic solutions, corresponding to distinct characteristic values of Fredholm.

Integral and its associate equation, are orthogonal.

3.3 Fundamental Functions

Let $D(\lambda)$ is the Fredholm's determinant for the kernel $K(x, \xi)$ and $D(\lambda_0) = 0$, then λ is said to be characteristic constant of the kernel. Suppose $\phi(x) \neq 0$, and is continuous in an interval (a, b) .

$$\phi(x) = \lambda_0 \int_a^b K(x, \xi) \phi(\xi) d\xi$$

Then $\phi(x)$ is called a fundamental function of the kernel $K(x, \xi)$ corresponding to the characteristic constant λ_0 .

Let the solution $\phi_1(x), \phi_2(x), \dots, \phi_n(x)$ which is expressible linearly constitute a complete system of fundamental function of the kernel $K(x, \xi)$ relating to λ_0 . Thus, if $\psi_1, \psi_2, \dots, \psi_n$ are any other n solutions, we have

$$\psi_1 = C_{11}\phi_1 + C_{12}\phi_2 + \dots + C_{1n}\phi_n,$$

$$\psi_2 = C_{21}\phi_1 + C_{22}\phi_2 + \dots + C_{2n}\phi_n$$

$$\dots \dots \dots$$

$$\dots \dots \dots$$

$$\psi_n = C_{n1}\phi_1 + C_{n2}\phi_2 + \dots + C_{nn}\phi_n,$$

$$\Rightarrow \begin{vmatrix} C_{11} & C_{12} & \dots & C_{1n} \\ C_{21} & C_{22} & \dots & C_{2n} \\ \dots & \dots & \dots & \dots \\ C_{n1} & C_{n2} & \dots & C_{nn} \end{vmatrix} = 0$$

Hence $\psi_1, \psi_2, \dots, \psi_n$ form a complete system of fundamental functions corresponding to λ_0 .

3.4 Integral Equations with Degenerate Kernels

The kernel $K(x, \xi)$ of Fredholm integral equation of the second kind is called degenerate (or separable) if it can be expressed as the sum of finite number of terms, each of which is the product of a function of x alone and a function of ξ alone i.e., if it is of the form.

$$K(x, \xi) = \sum_{v=1}^n a_v(x)b_v(\xi) \quad (1)$$

There is no loss in generality if we assume that n functions $a_v(x)$ and $b_v(\xi)$ are linearly independent and continuous in the basic square $a \leq x, \xi \leq b$ with (1) as its kernel, the Fredholm integral equation of second kind

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi \quad (2)$$

$$\text{Reduces to } \phi(x) = F(x) + \lambda \int_a^b \left[\sum_{v=1}^n a_v(x)b_v(\xi) \right] \phi(\xi) d\xi$$

$$\text{or } \phi(x) = F(x) + \lambda \sum_{v=1}^n a_v(x) \int_a^b b_v(\xi) \phi(\xi) d\xi \quad (3)$$

Assuming

$$C_v = \int_a^b b_v(\xi) \phi(\xi) d\xi, \quad (v=1, 2, \dots, n) \quad (4)$$

Substituting the relation (4) in the equation (3), we have

$$\phi(x) = F(x) + \lambda \sum_{v=1}^n C_v a_v(x) \quad (5)$$

Where C_v are unknown constants as the function $\phi(x)$ is unknown. Thus the equation (5) is the form of the required solution of the integral equation (2).

Multiplying both the sides of (5) successively by $b_1(x), b_2(x), \dots, b_n(x)$ and integrating over the interval (a, b) we have,

$$\int_a^b b_1(x) \phi(x) dx = \int_a^b b_1(x) F(x) dx + \lambda \int_a^b b_1(x) \sum_{v=1}^n C_v a_v(x) dx$$

$$\int_a^b b_2(x) \phi(x) dx = \int_a^b b_2(x) F(x) dx + \lambda \int_a^b b_2(x) \sum_{v=1}^n C_v a_v(x) dx$$

$$\int_a^b b_n(x) \phi(x) dx = \int_a^b b_n(x) F(x) dx + \lambda \int_a^b b_n(x) \sum_{i=1}^n C_i a_i(x) dx \quad (6)$$

Introducing the notation

$$\alpha_{mv} = \int_a^b b_m(x) a_v(x) dx, \beta_m = \int_a^b b_m(x) F(x) dx \quad (7)$$

The system of equations (6) reduce to

$$\begin{aligned} (1-\lambda\alpha_{11})C_1 - \lambda\alpha_{12}C_2 - \lambda\alpha_{13}C_3 - \dots - \lambda\alpha_{1n}C_n &= \beta_1 \\ -\lambda\alpha_{21}C_1 + (1-\lambda\alpha_{22})C_2 - \lambda\alpha_{23}C_3 - \dots - \lambda\alpha_{2n}C_n &= \beta_2 \end{aligned}$$

$$-\lambda\alpha_{n1}C_1 - \lambda\alpha_{n2}C_2 - \lambda\alpha_{n3}C_3 - \dots + (1 - \lambda\alpha_{nn})C_n = \beta_n \quad (8)$$

Equation (8) represent the linear algebraic system of n non-homogeneous equations for the n unknowns. The determinant $D(\lambda)$ of the system is

$$D(\lambda) = \begin{vmatrix} 1 - \lambda\alpha_{11} & -\lambda\alpha_{12} & \dots & -\lambda\alpha_{1n} \\ -\lambda\alpha_{21} & 1 - \lambda\alpha_{22} & \dots & -\lambda\alpha_{2n} \\ \dots & \dots & \dots & \dots \\ -\lambda\alpha_{n1} & -\lambda\alpha_{n2} & \dots & 1 - \lambda\alpha_{nn} \end{vmatrix} \quad (9)$$

Which is a polynomial in λ of degree at most n . The determinant $D(\lambda) \neq 0$, since, when $\lambda = 0$, it reduces to unity.

The system (8) may be written in the matrix notation as

$$(I - \lambda A)C = \beta$$

Where I is the unit matrix of order n and A is the matrix $[a_{ij}]_{n \times n}$. The set of equations possesses a unique solution for the C 's which is obtained from Cramer's rule if and only if the determinant $D(\lambda)$ of the coefficients of the C 's is non-singular in nature, i.e. $D(\lambda) \neq 0$

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$$C = \frac{1}{D(\lambda)} \begin{vmatrix} 1 - \lambda \alpha_{11} \dots - \lambda \alpha_{1v-1} \beta_1 & -\lambda \alpha_{1k+1} \dots - \lambda \alpha_{1n} \\ -\lambda \alpha_{21} \dots - \lambda \alpha_{2v-1} \beta_2 & -\lambda \alpha_{2k+1} \dots - \lambda \alpha_{2n} \\ \dots & \dots \\ -\lambda \alpha_{n1} \dots - \lambda \alpha_{nv-1} \beta_n & -\lambda \alpha_{nk+1} \dots 1 - \lambda \alpha_{nn} \end{vmatrix}$$

If the function $F(x) \equiv 0$, the integral equation reduces to the homogeneous equation, and is obviously satisfied by the trivial solution $\phi(x) \equiv 0$, corresponding to the trivial solution $C_1 = C_2 = \dots = C_n = 0$ of (8). However, if $|I - \lambda A| = 0$, at least one of the C 's can be assigned arbitrarily and the remaining C 's can be determined accordingly. In such cases, infinitely many solutions of the integral equation exist.

Those values of the parameter λ for which $D(\lambda) = 0$ are called eigen values or characteristic numbers of the homogeneous equation and every non-trivial solution of the homogeneous integral equation is called an eigen function or characteristic function corresponding to the eigen value λ . If K of the constant $C_1, C_2, \dots, C_n$ can be assigned arbitrarily for a given characteristic value of λ , then K linearly independent corresponding characteristic functions are obtained.

If the function $F(x) \neq 0$ but is orthogonal to all the functions $b_1(x), b_2(x), \dots, b_n(x)$ then the equation (7) shows that the right hand side of (8) again vanish, and the trivial solution $C_1 = C_2 = \dots = C_n = 0$ lead to the solution $\phi = F(x)$. Thus the solution corresponding to eighevalue λ are expressed as the sum of $F(x)$ and arbitrary multiples of eigen functions.

If at least one of the members on R.H.S. of (8) does not vanish then a unique non-trivial solution exists, when the determinant $D(\lambda) \neq 0$, the system of equations (8) are either incompatible and no solution exists or they are redundant, and infinitely many solutions exist.

Example: Solve the following integral equation

$$\phi(x) = x + \lambda \int_{-\pi}^{\pi} (x \cos \xi + \xi^2 \sin x + \cos x \sin \xi) \phi(\xi) d\xi$$

$$\text{Solution: } \phi(x) = x + \lambda \int_{-\pi}^{\pi} (x \cos \xi + \xi^2 \sin x + \cos x \sin \xi) \phi(\xi) d\xi$$

Writing the integral equation in the following from:

$$\phi(x) = x + \lambda \left[x \int_{-\pi}^{\pi} \cos \xi \phi(\xi) d\xi + \sin x \int_{-\pi}^{\pi} \xi^2 \phi(\xi) d\xi + \cos x \int_{-\pi}^{\pi} \sin \xi \phi(\xi) d\xi \right] \quad (1)$$

$$\text{or } \phi(x) = x + \lambda [C_1 x + C_2 \sin x + C_3 \cos x],$$

$$\text{Where } C_1 = \int_{-\pi}^{\pi} \cos \xi \phi(\xi) d\xi, \quad C_2 = \int_{-\pi}^{\pi} \xi^2 \phi(\xi) d\xi,$$

$$C_3 = \int_{-\pi}^{\pi} \sin \xi \phi(\xi) d\xi \quad (2)$$

Substituting expression (1) into (2), we obtain

$$C_1 = \int_{-\pi}^{\pi} \cos \xi [\xi + C_1 \lambda \xi + C_2 \lambda \sin \xi + C_3 \lambda \cos \xi] d\xi$$

$$\text{or } C_2 = \int_{-\pi}^{\pi} \xi^2 [\xi + C_1 \lambda \xi + C_2 \lambda \sin \xi + C_3 \lambda \cos \xi] d\xi$$

$$\text{or } C_3 = \int_{-\pi}^{\pi} \sin \xi [\xi + C_1 \lambda \xi + C_2 \lambda \sin \xi + C_3 \lambda \cos \xi] d\xi$$

$$\text{or } C_1 \left[1 - \lambda \int_{-\pi}^{\pi} \xi \cos \xi d\xi \right] - C_2 \lambda \int_{-\pi}^{\pi} \sin \xi \cos \xi d\xi - C_3 \lambda \int_{-\pi}^{\pi} \cos^2 \xi d\xi = \int_{-\pi}^{\pi} \xi \cos \xi d\xi$$

$$\text{Or } -C_1 \lambda \int_{-\pi}^{\pi} \xi^3 d\xi + C_2 \left[1 - \lambda \int_{-\pi}^{\pi} \xi^2 \sin \xi d\xi \right] - C_3 \lambda \int_{-\pi}^{\pi} \xi^2 \cos \xi d\xi = \int_{-\pi}^{\pi} \xi^3 d\xi$$

$$\text{Or } -C_1 \lambda \int_{-\pi}^{\pi} \xi \sin \xi d\xi - C_2 \lambda \int_{-\pi}^{\pi} \sin^2 \xi d\xi + C_3 \int_{-\pi}^{\pi} \left[1 - \lambda \int_{-\pi}^{\pi} \cos \xi \sin \xi d\xi \right] = \int_{-\pi}^{\pi} \xi \sin \xi d\xi$$

By evaluating the integrals, we obtain a system of algebraic equations

$$C_1 + 0C_2 - \lambda\pi C_3 = 0$$

$$C_1 + C_2 + 4\lambda\pi C_3 = 0$$

$$-2\lambda\pi C_1 - \lambda\pi C_2 + C_3 = 2\pi \quad (3)$$

The determinant of this system is

$$D(\lambda) = \begin{vmatrix} 1 & 0 & -\lambda\pi \\ 0 & 1 & 4\lambda\pi \\ -2\lambda\pi & -\lambda\pi & 1 \end{vmatrix} = 1 + 2\lambda^2\pi^2 \neq 0$$

Thus the system (3) has a unique solution.

$$C_1 = \frac{2\lambda\pi^2}{1+2\lambda^2\pi^2}, C_2 = \frac{+8\lambda\pi^2}{1+2\lambda^2\pi^2}, C_3 = \frac{2\pi}{1+2\lambda^2\pi^2}$$

Substituting the value of C_1, C_2 and C_3 in (1), we obtain the solution of given integral equation as follows

$$\phi(x) = x + \frac{2\lambda\pi}{1+2\lambda^2\pi^2} [\lambda\pi x - 4\lambda\pi \sin x + \cos x]$$

Example: Solve the following integral equation

$$\phi(x) = x + \lambda \int_0^{\pi} (1 + \sin x \sin \xi) \phi(\xi) d\xi$$

$$\text{Solution: } \phi(x) = x + \lambda \int_0^{\pi} (1 + \sin x \sin \xi) \phi(\xi) d\xi \quad (1)$$

Writing the integral equation in the following form

$$\phi(x) = x + \lambda \left[\int_0^{\pi} \phi(\xi) d\xi + \sin x \int_0^{\pi} \sin \xi \phi(\xi) d\xi \right]$$

$$\text{or } \phi(x) = x + \lambda [C_1 + C_2 \sin x]$$

$$\text{Where } C_1 = \int_0^{\pi} \phi(\xi) d\xi, C_2 = \int_0^{\pi} \sin \xi \phi(\xi) d\xi \quad (2)$$

Substituting expression (1) into (2), we have

$$C_1 = \int_0^\pi (\xi + \lambda C_1 + \lambda C_2 \sin \xi) d\xi$$

$$C_2 = \int_0^\pi \sin \xi (\xi + \lambda C_1 + \lambda C_2 \sin \xi) d\xi$$

$$\text{Or } C_1 \left[1 - \lambda \int_0^\pi d\xi \right] - C_2 \lambda \int_0^\pi \sin \xi d\xi = \int_0^\pi \xi d\xi$$

$$\text{Or } -C_1 \lambda \int_0^\pi \sin \xi d\xi + C_2 \left[1 - \lambda \int_0^\pi \sin^2 \xi d\xi \right] = \int_0^\pi \xi \sin \xi d\xi$$

By evaluating the integrals, we obtain a system of algebraic equations.

$$(1 - \lambda\pi)C_1 - 2\lambda C_2 = \frac{\pi^2}{2}$$

$$-2\lambda C_1 + \left(1 - \frac{\lambda\pi}{2}\right)C_2 = \pi \quad (3)$$

The determinant of this system (3) is given by

$$\begin{vmatrix} 1 - \lambda\pi & -2\lambda \\ -2\lambda & 1 - \frac{\lambda\pi}{2} \end{vmatrix} = (1 - \lambda\pi) \left(1 - \frac{\lambda\pi}{2}\right) + 4\lambda^2 \neq 0$$

The system (3) has a unique solution

$$C_1 = \frac{2\lambda\pi + \frac{\pi^2}{2} \left(1 - \frac{\lambda\pi}{2}\right)}{(1 - \lambda\pi) \left(1 - \frac{\lambda\pi}{2}\right) - 4\lambda^2}$$

$$C_2 = \frac{\pi(1 - 2\lambda\pi)}{(1 - \lambda\pi) \left(1 - \frac{\lambda\pi}{2}\right) - 4\lambda^2}$$

Substituting the values of C_1 and C_2 in (1), we obtain the solution of the integral equation as

$$\phi(x) = x + \lambda \left[\frac{2\lambda\pi + \frac{\pi^2}{2} \left(1 - \frac{\lambda\pi}{2}\right)}{(1 - \lambda\pi) \left(1 - \frac{\lambda\pi}{2}\right) - 4\lambda^2} + \frac{\pi(1 - 2\lambda\pi)}{(1 - \lambda\pi) \left(1 - \frac{\lambda\pi}{2}\right) - 4\lambda^2} \sin x \right]$$

$$\text{or } \phi(x) = x + \frac{\lambda}{(1 - \lambda\pi) \left(1 - \frac{\lambda\pi}{2}\right) - 4\lambda^2}$$

$$\left[\left\{ 2\lambda\pi + \frac{1}{2}\pi^2(1 - \frac{1}{2}\lambda\pi) \right\} + \pi(1 - 2\lambda\pi)\sin x \right]$$

Example: Solve the integral equation

$$\phi(x) = 1 + \int_0^1 (1 + e^{x+\xi}) \phi(\xi) d\xi$$

$$\text{Solution: } \phi(x) = 1 + \int_0^1 (1 + e^{x+\xi}) \phi(\xi) d\xi \quad (1)$$

$$\text{or } \phi(x) = 1 + \int_0^1 \phi(\xi) d\xi + e^x \int_0^1 e^\xi \phi(\xi) d\xi$$

$$\text{or } \phi(x) = 1 + C_1 + C_2 e^x. \quad (2)$$

$$\text{where } C_1 = \int_0^1 \phi(\xi) d\xi \text{ and } C_2 = \int_0^1 e^\xi \phi(\xi) d\xi \quad (3)$$

From the equation (2), the relation (3) reduces to

$$C_1 = \int_0^1 (1 + C_1 + C_2 e^\xi) d\xi \Rightarrow C_1 = 1 + C_1 + C_2(e-1)$$

$$\text{and } C_2 = \int_0^1 e^\xi (1 + C_1 + C_2 e^\xi) d\xi$$

$$\Rightarrow C_2 = (1 + C_1)(e-1) + \frac{1}{2} C_2 (e^2 - 1)$$

$$\Rightarrow C_2 = \frac{-1}{e-1} \text{ and } C_1(e-1) = C_2 - (e-1) - \frac{1}{2} C_2 (e^2 - 1)$$

$$\text{or } C_1(e-1) = -\frac{1}{e-1} - (e-1) + \frac{1}{2}(e+1) = -\frac{1}{e-1} - \frac{1}{2}(e-3)$$

or $C_1 = -\frac{e^2 - 4e + 5}{2(e-1)^2}$ Substituting the value of C_1 and C_2 in equation (2), we have,

$$\phi(x) = 1 - \frac{e^2 - 4e + 5}{2(e-1)^2} - \frac{1}{(e-1)} e^x$$

$$\text{or } \phi(x) = \frac{(e^2 - 3) - 2(e-1)e^x}{2(e-1)^2}$$

Example: Prove that the integral equation

$$\phi(x) = \lambda \int_0^1 [\sqrt{(x)} \xi - \sqrt{(\xi)} x] \phi(\xi) d\xi$$

does not have real characteristic numbers and characteristic functions.

$$\text{Solutions: } \phi(x) = \lambda \left[\sqrt{(x)} \int_0^1 \xi \phi(\xi) d\xi - x \int_0^1 \sqrt{(\xi)} \phi(\xi) d\xi \right]$$

$$\text{or } \phi(x) = \lambda [C_1 \sqrt{(x)} - C_2 x] \quad (1)$$

$$\text{Where } C_1 = \int_0^1 \xi \phi(\xi) d\xi \text{ and } C_2 = \int_0^1 \sqrt{(\xi)} \phi(\xi) d\xi \quad (2)$$

Using the relation (1) into (2), we obtain

$$C_1 = \int_0^1 \lambda \xi (C_1 \sqrt{(\xi)} - C_2 \xi) d\xi$$

$$\text{And } C_2 = \int_0^1 \lambda \sqrt{\xi} (C_1 \sqrt{\xi} - C_2 \xi) d\xi$$

$$\text{or } \left[1 - \lambda \int_0^1 \xi^{3/2} d\xi \right] C_1 + \left[\lambda \int_0^1 \xi^2 d\xi \right] C_2 = 0$$

$$\text{and } \left[-\lambda \int_0^1 \xi d\xi \right] C_1 + \left[1 + \lambda \int_0^1 \xi^{3/2} d\xi \right] C_2 = 0$$

$$\left(1 - \frac{2\lambda}{5} \right) C_1 + \frac{\lambda}{3} C_2 = 0$$

$$-\frac{\lambda}{2} C_1 + \left(1 + \frac{2\lambda}{5} \right) C_2 = 0 \quad (3)$$

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The determinant of the system is given by

$$D(\lambda) = \begin{vmatrix} 1 - \frac{2\lambda}{5} & \frac{\lambda}{3} \\ -\frac{\lambda}{2} & 1 + \frac{2\lambda}{5} \end{vmatrix} = 1 + \frac{\lambda^2}{150}$$

This does not vanish for real λ , so from (3), we obtain

$$C_1 = 0 \text{ and } C_2 = 0$$

$\Rightarrow$ That the algebraic equations contain only one solution for all real λ , i.e., the trivial solution

$\phi(x) \equiv 0$, therefore the integral equation does not have real characteristic numbers of characteristic functions.

EXERCISE (3)

1. Using Fredholm determinants, find the resolvent kernel of the following kernels.

- (a) $K(x, \xi) = 2x - \xi$; $0 \leq x \leq 1, 0 \leq \xi \leq 1$
 (b) $K(x, \xi) = x^2\xi - x\xi^2$; $0 \leq x \leq 1, 0 \leq \xi \leq 1$
 (c) $K(x, \xi) = \sin x - \sin \xi$; $0 \leq x \leq 2\pi; 0 \leq \xi \leq 2\pi$
 (d) $K(x, \xi) = 1 + 3x\xi$; $0 \leq x \leq 1, 0 \leq \xi \leq 1$

2. Solve the following integral equations

- (a) $\phi(x) = \sec^2 x + \lambda \int_0^1 \phi(\xi) d\xi$
 (b) $\phi(x) = e^x + \lambda \int_0^{10} x\xi \phi(\xi) d\xi$
 (c) $\phi(x) = \sin x + \lambda \int_4^{10} x\phi(\xi) d\xi$

3. Solve the following integral equations.

- (a) $\phi(x) = \cos x + \lambda \int_0^\pi \sin x \phi(\xi) d\xi$
 (b) $\phi(x) = (2x - \pi) + 4 \int_0^{\frac{\pi}{2}} \sin^2 x \phi(\xi) d\xi$
 (c) $\phi(x) = x + \lambda \int_0^1 (1 + x + \xi) \phi(\xi) d\xi$
 (d) $\phi(x) = x + \lambda \int_0^1 (x\xi^2 + x^2\xi) \phi(\xi) d\xi$
 (e) $\phi(x) = \left(\frac{6}{5}\right)(1 - 4x) + \lambda \int_0^1 (x \log \xi - \xi \log x) \phi(\xi) d\xi$

4. Determine the eigen values and eigen function for the following homogeneous equations with degenerate kernels.

- (a) $\phi(x) = \lambda \int_{-1}^1 (5x\xi^2 + 4x^2\xi + 3x\xi) \phi(\xi) d\xi$
 (b) $\phi(x) = \lambda \int_0^1 (2x\xi - 4x^2) \phi(\xi) d\xi$
 (c) $\phi(x) = \frac{1}{e^2 - 1} \int_0^1 2e^x e^\xi \phi(\xi) d\xi$

Put Your Own Notes

CHAPTER 4

HILBERT SCHMIDT THEORY

4.1. Definition

The theory of Hilbert and Schmidt dealt with real symmetric kernels satisfying the conditions.

$$K(x, \xi) = K(\xi, x)$$

The more general case of Hermitian kernels are defined as $K(x, \xi) = K(\xi, x)$, Where $K(x, \xi)$ is not necessarily real.

Thus the kernel $K(x, \xi)$ of a Fredholm integral equation of the second kind.

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi,$$

is called symmetric (or Hermitian) if it is identical to the kernel of the associated equation, i.e., if $K(x, \xi) = K(\xi, x)$,

Where $K(\xi, x)$ denotes the complex conjugate. The integral operator

$K = \int_a^b K(x, \xi) d\xi$ is Hermitian or self adjoint since it satisfies the condition

$(K\phi, \psi) = (\phi, K\psi)$ for all ϕ, ψ belonging to the Hilbert space of L^2 function.

$$\Rightarrow (K\psi, \psi) = (\psi, K\psi) = \sqrt{(K\psi, \psi)}$$

A Hermitian kernel which is square integral is called a Hilbert Schmidt kernel $K(x, \xi)$.

4.2. Theorem

All iterated kernels of a symmetric kernel are also symmetric.

Proof: The iterated kernel is defined as $K_1(x, \xi) = K(x, \xi)$,

$$\text{or } K_2(x, \xi) = \int_a^b K(x, z) K_1(z, \xi) dz \Rightarrow \int_a^b K(x, z) \cdot K(z, y) dz$$

$$K_2(\xi, x) = \int_a^b \overline{K(\xi, z)} \overline{K(z, x)} dz$$

The recursion relation gives $K_n(x, \xi) = \int_a^b K_1(x, z) K_{n-1}(z, \xi) dz$

$$\Rightarrow K_{n+m}(x, \xi) = \int_a^b K_n(x, z) K_m(z, \xi) dz$$

$$\text{or } K_n(\xi, x) = \int_a^b \int_a^b K(\xi, z_1) K(z_1, z_2) \dots K(z_{n-1}, x) dz_1, dz_2, \dots, dz_{n-1}$$

$$= \int_a^b \int_a^b K(z_1, \xi) K(z_2, z_1) \dots K(x, z_{n-1}) dz_1 dz_2 \dots dz_{n-1}$$

Since $K(x, \xi)$ is symmetric, it reduces to

$$= \int_a^b \int_a^b K(x, z_{n-1}) \dots K(z_2, z_1) K(z_1, \xi) dz_1 dz_2 \dots dz_{n-1}.$$

Substituting $z_{n-1}, z_{n-2}, \dots, z_2, z_1$ for $z_1, z_2, \dots, z_{n-2}, z_{n-1}$, we have

$$K_n(\xi, x) = \int_a^b \int_a^b K(x, z_1) K(z_1, z_2) \dots K(z_{n-1}, \xi) dz_1 dz_2 \dots dz_{n-1}.$$

$$\text{or } K_n(\xi, x) = K_n(x, \xi)$$

Thus the kernel K_n is also symmetric. Hence, by induction, we conclude that every iterated kernel of a symmetric kernel is symmetric.

The above relation implies that $K_n(x, x) = \bar{K}_n(x, x)$ and therefore the function $K_n(x, x)$ are real.

4.3. Orthogonality

Two functions $\phi_1(x)$ and $\phi_2(x)$ continuous on an interval (a, b) are said to be orthogonal if.

$$(\phi_1, \phi_2) = \int_a^b \phi_1(x) \phi_2(x) dx = 0$$

Theorem

If $K(x, \xi)$ is symmetric and $\phi_m(x)$ and $\phi_n(x)$ are fundamental functions of $K(x, \xi)$ for λ_m and λ_n respectively ($\lambda_m \neq \lambda_n$) then $\phi_m(x)$ and $\phi_n(x)$ are orthogonal on an interval (a, b) i.e. $\int_a^b \phi_m(x) \phi_n(x) dx = 0$

Proof: Let $\phi_m(x)$ and $\phi_n(x)$ be the eigen functions corresponding to two eigen values λ_m and λ_n

Respectively, of the homogeneous Fredholm integral equation.

$$\phi(x) = \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi \quad (1)$$

and the kernel $K(x, \xi)$ is symmetric so that

$$K(x, \xi) = K(\xi, x) \quad (2)$$

We notice that $\lambda = 0$ cannot be an eigen value since it gives the trivial solution $\phi(x) = 0$. The function $\phi_m(x)$ and $\phi_n(x)$ must satisfy the integral equation (1), then

$$\phi_m(x) = \lambda_m \int_a^b K(x, \xi) \phi_m(\xi) d\xi \quad (3)$$

$$\phi_n(x) = \lambda_n \int_a^b K(x, \xi) \phi_n(\xi) d\xi \quad (4)$$

Multiplying (3) by $\phi_n(x)$ both the sides and integrating with regard to x over an interval (a, b) .

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$\int_a^b \phi_m(x) \phi_n(x) dx = \lambda_m \int_a^b \phi_n(x) \left\{ \int_a^b K(x, \xi) \phi_m(\xi) d\xi \right\} dx$ changing the order of integration on R.H.S. we have

$$\int_a^b \phi_m(x) \phi_n(x) dx = \lambda_m \int_a^b \phi_m(\xi) \left\{ \int_a^b K(x, \xi) \phi_n(x) d\xi \right\} d\xi$$

Since the kernel $K(x, \xi)$ is symmetric, thus rewriting the inner integral on the R.H.S. in the form

$$\int_a^b \phi_m(x) \phi_n(x) dx = \lambda_m \int_a^b \phi_m(\xi) \left\{ \int_a^b K(\xi, x) \phi_n(x) dx \right\} d\xi \quad (5)$$

By interchanging the variables x and ξ in (4), we have

$$\phi_n(\xi) = \lambda_n \int_a^b K(\xi, x) \phi_n(x) dx \Rightarrow \int_a^b K(\xi, x) \phi_n(x) dx = \frac{\phi_n(\xi)}{\lambda_n} \quad (6)$$

Using the relation (6), the equation (5) reduces to

$$\int_a^b \phi_m(x) \phi_n(x) dx = \frac{\lambda_m}{\lambda_n} \int_a^b \phi_m(\xi) \phi_n(\xi) d\xi$$

$$\text{or } \left(1 - \frac{\lambda_m}{\lambda_n} \right) \int_a^b \phi_m(x) \phi_n(x) dx = 0$$

$$\text{or } \int_a^b \phi_m(x) \phi_n(x) dx = 0, \text{ since } \lambda_m \neq \lambda_n \quad (7)$$

Thus the conclusion that if $\phi_m(x)$ and $\phi_n(x)$ are eigen functions of (1) corresponding to distinct eigen values then $\phi_m(x)$ and $\phi_n(x)$ are orthogonal over the interval (a, b) .

Theorem

The eigen values of a symmetric kernel are real.

Proof: Let $\lambda_m = \alpha + i\beta$ ($\beta \neq 0$) is a complex eigen value corresponding to an eigen function $\phi_m(x) = \psi_1 + i\psi_2$ then the complex conjugate number $\bar{\lambda}_m = \alpha - i\beta$ must be an eigen value corresponding to the eigen function $\bar{\phi}_m(x) = \bar{\psi}_1 - i\bar{\psi}_2$. Thus by replacing λ_n by $\bar{\lambda}_m$ and $\phi_n(x)$ by $\bar{\phi}_m(x)$ in (7), it follows that

$$(\lambda_m - \bar{\lambda}_m) \int_a^b \phi_m(x) \bar{\phi}_m(x) dx = 0$$

$$(\alpha + i\beta - \alpha + i\beta) \int_a^b (\psi_1 + i\psi_2) (\bar{\psi}_1 - i\bar{\psi}_2) dx = 0$$

$$(2i\beta) \int_a^b (\psi_1 \bar{\psi}_1 + \psi_2 \bar{\psi}_2) dx = 0$$

$$\text{or } (2i\beta) \int_a^b (\psi_1^2 + \psi_2^2) dx = 0$$

Since $\phi_m(x) \neq 0$, so the integral cannot vanish. Thus the imaginary part of λ_m must vanish, i.e., $\beta = 0$. Hence the eigen values of a symmetric kernel are real.

Theorem

If $K(x, \xi)$ is real and symmetric continuous and identically not equal to zero then all of the characteristic constants are real.

Proof: Let the characteristic constant λ_0 is not real, then

$$\lambda_0 = \mu_0 + i\nu_0, \nu_0 \neq 0$$

The homogeneous integral equation

$$\phi(x) = \lambda_0 \int_a^b K(x, \xi) \phi(\xi) d\xi$$

has at least one continuous solution $u(x) \equiv 0$ (1)

$$u(x) = (\mu_0 + i\nu_0) \int_a^b K(x, \xi) u(\xi) d\xi \quad (2)$$

separating into real and imaginary parts, considering $\bar{u}(x)$ as real.

$$u(x) = \mu_0 \int_a^b K(x, \xi) u(\xi) d\xi \quad (3)$$

$$0 = \nu_0 \int_a^b K(x, \xi) u(\xi) d\xi \quad (4)$$

From (4), we get

$$\int_a^b K(x, \xi) u(\xi) d\xi = 0 \text{ since } \nu_0 \neq 0$$

Thus, we have

$$u(x) \equiv 0$$

Which is contrary to our assumption $u(x) \equiv 0$. Hence $u(x)$ cannot be real

Let $u(x) = p(x) + iq(x)$, where $p(x)$ and $q(x)$ are real.

From the equation (2), we obtain

$$p(x) + iq(x) = (\mu_0 + i\nu_0) \int_a^b K(x, \xi) u(\xi) d\xi \quad (5)$$

Separating into real and imaginary parts, we have

$$p(x) = \mu_0 \int_a^b K(x, \xi) p(\xi) d\xi - \nu_0 \int_a^b K(x, \xi) q(\xi) d\xi \quad (6)$$

$$q(x) = \mu_0 \int_a^b K(x, \xi) q(\xi) d\xi - \nu_0 \int_a^b K(x, \xi) p(\xi) d\xi \quad (7)$$

Multiplying (7) both the sides with $-i$, and adding it to (6), we get

$$p(x) - iq(x) = (\mu_0 - i\nu_0) \int_a^b K(x, \xi) [p(\xi) - iq(\xi)] d\xi$$

Thus $p(x) - iq(x) \{\equiv \bar{u}(x)\}$ is also a fundamental function of $K(x, \xi)$ relating to $\bar{\lambda}_0 = \mu_0 - i\nu_0$. Since $\lambda_0 \neq \bar{\lambda}_0$ then by orthogonal property, we have

$$\int_a^b u(x) \bar{u}(x) dx = 0$$

$$\text{or } \int_a^b [p^2(x) + q^2(x)] dx = 0$$

But $p(x)$ and $q(x)$ are real functions. Therefore

$$p^2(x) + q^2(x) = 0 \Rightarrow p(x) \equiv 0, q(x) \equiv 0.$$

So $u(x) \equiv 0$ which is also contrary to our assumption. Therefore λ_0 cannot be of the form $\lambda_0 = \mu_0 + i\nu_0 (\nu_0 \neq 0)$ and hence λ_0 must be real.

4.4. Schmidt's solution of the non-homogeneous integral equation (when λ is not a characteristic constant).

Consider a non-homogeneous Fredholm integral equation of second kind as

$$\phi(x) = F(x) + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi \quad (1)$$

let $\phi(x)$ be a continuous solution of (1), then

$$\phi(x) - F(x) = \int_a^b K(x, \xi) \{\lambda \phi(\xi)\} d\xi = \psi(x) \text{ (say)} \quad (2)$$

If the equation (1) possesses continuous solution $\phi(x)$ then the function $[\phi(x) - F(x)]$ is generated by (2), which can be represented by a linear combination of the normalized characteristic functions $\psi_n(x)$ of the form

$$\phi(x) - F(x) = \sum a_n \psi_n(x), a \leq x \leq b \quad (3)$$

$$\text{or } a_n = \int_a^b [\phi(x) - F(x)] \psi_n(x) dx$$

$$\text{or } a_n = \int_a^b \phi(x) \psi_n(x) dx - \int_a^b F(x) \psi_n(x) dx$$

$$\text{or } a_n = C_n - f_n,$$

$$\text{Where } C_n = \int_a^b \phi(x) \psi_n(x) dx, f_n = \int_a^b F(x) \psi_n(x) dx \quad (5)$$

Multiplying (1) by $\psi_n(x)$ and integrating with regard to x over (a, b) , we have

$$\int_a^b \phi(x) \psi_n(x) dx = \int_a^b F(x) \psi_n(x) dx + \lambda \int_a^b \psi_n(x) dx \left\{ \int_a^b K(x, \xi) \phi(\xi) d\xi \right\}$$

By changing the order of integration of the second term on R.H.S, we have

$$C_n = f_n + \lambda \int_a^b \phi(\xi) \left\{ \int_a^b K(\xi, x) \psi_n(x) dx \right\} d\xi$$

$$\text{or } C_n = f_n + \frac{\lambda}{\lambda_n} \int_a^b \phi(\xi) \psi_n(\xi) d\xi$$

$$\text{or } C_n = f_n + \frac{\lambda}{\lambda_n} C_n \Rightarrow C_n = \left(\frac{\lambda_n}{\lambda_n - \lambda} \right) f_n \quad (6)$$

From (4) and (6) we have

$$a_n = \left(\frac{\lambda_n}{\lambda_n - \lambda} \right) f_n = \left(\frac{\lambda}{\lambda_n - \lambda} \right) f_n \quad (7)$$

Thus the required solution of the integral equation (1) is given by (3), determined as

$$\phi(x) - F(x) = \lambda \sum_{n=1}^{\infty} \frac{f_n}{\lambda_n - \lambda} \psi_n(x)$$

or $\phi(x) = F(x) + \lambda \sum_{n=1}^{\infty} \frac{f_n}{\lambda_n - \lambda} \psi_n(x) \quad (8)$

Where $\lambda \neq \lambda_n$, f_n is defined by the relation (6) and $\psi_n(x)$ are the characteristic functions of the corresponding homogeneous integral equation of (1). The series on the R.H.S. of formula (8) converges absolutely and uniformly in the square $a \leq x, \xi \leq b$, The solution (8) exists uniquely of λ does not take a characteristic value. If $\lambda = \lambda_m$, where λ_m is the m th characteristic number, the solution (8) does not exist unless also $f_m = 0$, i.e., unless $F(x)$ is orthogonal to the corresponding characteristic function. But if $\lambda = \lambda_m$ and $f_m = 0$, the relation (6) reduces to a trivial identity. It follows that the coefficient of $\psi_n(x)$ in (8) are arbitrary hence posses infinitely distinct many solutions. If λ takes a characteristic value and $F(x)$ is not orthogonal to the corresponding characteristic function then no solution exists.

On the other hand, if the parameter λ coincides with one of the characteristic number, say $\lambda = \lambda_m$, of index q , then the equation(1) will not contain any solutions. The solution exist iff the m conditions are satisfied

$$\int_a^b F(x) \psi_m(x) dx = 0, (m = 1, 2, 3, \dots, q) \quad (9)$$

It follows that the function $F(x)$ is orthogonal to all characteristic function belonging to the characteristic number λ_m . The equation (1) has an infinite solutions which contain arbitrary constants and are given by

$$\phi(x) = F(x) + \lambda \sum_{n=q+1}^{\infty} \psi_n(x) + C_1 \psi_1(x) + C_2 \psi_2(x) + \dots + C_q \psi_q(x) \quad (1)$$

Where $C_1, C_2, \dots, C_q$ are arbitrary constants.

Example: Solve the integral equation

$$\phi(x) = e^x + \lambda \int_a^b K(x, \xi) \phi(\xi) d\xi$$

$$\text{Where } K(x, \xi) = \begin{cases} \frac{\sinh x \sinh(\xi - 1)}{\sinh 1}, & 0 \leq x \leq \xi \\ \frac{\sinh \xi \sinh(x - 1)}{\sinh 1}, & \xi \leq x \leq 1 \end{cases}$$

Put Your Own Notes

Solution: Consider the homogeneous integral equation in the form

$$\phi(x) = \lambda \left\{ \int_0^x K(x, \xi) \phi(\xi) d\xi + \int_x^1 K(x, \xi) \phi(\xi) d\xi \right\}$$

$$\phi(x) = \frac{\lambda \sinh(x-1)}{\sinh 1} \int_0^x \sinh \xi \phi(\xi) d\xi + \frac{\lambda \sinh x}{\sinh 1} \int_x^1 \sinh(\xi-1) \phi(\xi) d\xi \quad (1)$$

Differentiating (1), two times, with regard to x , we have

$$\phi'(x) = \frac{\lambda \cosh(x-1)}{\sinh 1} \int_0^x \sinh \xi \phi(\xi) d\xi + \frac{\lambda \sinh(x-1)}{\sinh 1} \sinh x \phi(x) + \frac{\lambda \cosh x}{\sinh 1} \int_x^1 \sinh(\xi-1) \phi(\xi) d\xi$$

$$\phi(\xi) d\xi - \frac{\lambda \sinh x}{\sinh 1} \sinh(x-1) \phi(x)$$

$$\text{or } \phi'(x) = \frac{\lambda \cosh(x-1)}{\sinh 1} \int_0^x \sinh \xi \phi(\xi) d\xi + \frac{\lambda \cosh x}{\sinh 1} \int_x^1 \sinh(\xi-1) \phi(\xi) d\xi$$

or

$$\phi''(x) = \frac{\lambda \cosh(x-1)}{\sinh 1} \int_0^x \sinh \xi \phi(\xi) d\xi + \frac{\lambda \sinh x}{\sinh 1} \int_x^1 \sinh(\xi-1) \phi(\xi) d\xi + \frac{\lambda \cosh(x-1)}{\sinh 1} \sinh x \phi(x)$$

$$- \frac{\lambda \cosh x}{\sinh 1} \sinh(x-1) \phi(x).$$

$$\text{or } \phi''(x) = \phi(x) + \frac{\lambda \phi(x)}{\sinh 1} \{ \cosh(x-1) \sinh x - \cosh x \sinh(x-1) \}$$

$$\text{or } \phi''(x) = \phi(x) + \frac{\lambda \phi(x)}{\sinh 1} \sinh[x(x-1)]$$

$$\text{or } \phi''(x) = \phi(x) + \lambda \phi(x)$$

$$\text{or } \phi''(x) - (\lambda + 1) \phi(x) = 0 \quad (2)$$

From (1), we have

$$\phi(0) = 0, \phi(1) = 0 \quad (3)$$

The required function $\phi(x)$ is a solution of the non-homogeneous boundary value problem (2) together with (3).

Case I: If $\lambda + 1 = 0, \Rightarrow \lambda = -1$, then equation (2) takes the form $\phi''(x) = 0$,

$$\text{Whose solution is given by } \phi(x) = Ax + B \quad (4)$$

From (3) and (4), we have

$$B = 0 \text{ and } A + B = 0 \Rightarrow A = 0$$

Thus we have $\phi(x) \equiv 0$ which is not an eigen function. Hence $\lambda = -1$ is not an Eigen value.

Case II: $\lambda + 1 > 0$, then equation (2) takes the form

$$\phi''(x) - (\lambda + 1)\phi(x) = 0$$

Whose general solution is given by

$$\phi(x) = A \cosh x \sqrt{(\lambda + 1)} + B \sinh \sqrt{(\lambda + 1)} x$$

From the boundary condition (3), we have

$$A = 0 \text{ and } A \cosh \sqrt{(\lambda + 1)} + B \sinh \sqrt{(\lambda + 1)} = 0$$

$$\text{or } \Rightarrow A = 0, B = 0$$

Thus, we have $\phi(x) \equiv 0$

Which is not an eigen function. Hence $\lambda + 1 > 0$ is not an eigen value.

Case III: If $\lambda + 1 < 0$, then $1 + \lambda = -\mu^2 = -n^2\pi^2 \Rightarrow \lambda = -(1 + n^2\pi^2)$

The eigen value λ_n are given by

$$\lambda_n = -(1 + n^2\pi^2), n = 1, 2, 3, \dots$$

Thus, we have

$$\phi(x) = B \sin n \pi x$$

Let $B = 1$, the required eigen function $\phi_n(x)$ are given as

$$\phi_n(x) = \sin n \pi x$$

The normalized eigen functions Φ_n are given by

$$\Phi_n(x) = \frac{\phi_n(x)}{\left[\int_0^1 [\phi_n(x)]^2 dx \right]^{1/2}} = \frac{\sin n \pi x}{\left[\int_0^1 \sin^2 n \pi x dx \right]^{1/2}}$$

$$\text{or } \Phi_n(x) = \frac{\sin n \pi x}{\left[\frac{1}{2} \int_0^1 (1 - \cos 2n \pi x) dx \right]^{1/2}} = \sqrt{2} \sin n \pi x.$$

Here $F(x) = e^x$

$$\text{Now } F_n = \int_0^1 F(x) \Phi_n(x) dx$$

$$\Rightarrow F_n = \int_0^1 e^x \cdot (\sqrt{2} \sin n \pi x) dx$$

$$\Rightarrow F_n = \left[\frac{\sqrt{2} e^x}{1 + n^2 \pi^2} \{ \sin n \pi x - n \pi \cos n \pi x \} \right]_0^1$$

$$\Rightarrow F_n = \frac{\sqrt{2}}{1 + n^2 \pi^2} \{ -n \pi e \cos n \pi + n \pi \}$$

$$\Rightarrow F_n = \frac{n \pi \sqrt{2} \{ 1 - e(-1)^n \}}{1 + n^2 \pi^2}, n = 1, 2, 3, \dots,$$

When λ be not an eigen value then the given integral equation will possess unique solution.

$$\begin{aligned}\phi(x) &= F(x) + \lambda \sum_{n=1}^{\infty} \frac{F_n}{\lambda_n - \lambda} \Phi_n(x), \lambda \neq \lambda_n \\ \Rightarrow \phi(x) &= e^x + \lambda \sum_{n=1}^{\infty} \frac{n\pi\sqrt{2}\{1 - e(-1)^n\}}{1 + n^2\pi^2} \cdot \frac{\sqrt{2}\sin n\pi x}{-n^2\pi^2 - 1 - \lambda} \\ \Rightarrow \phi(x) &= e^x - 2\lambda\pi \sum_{n=1}^{\infty} \frac{n\{1 - e(-1)^n\}\sin n\pi x}{(1 + n^2\pi^2)(1 + n^2\pi^2 + \lambda)}\end{aligned}$$

When λ be not an eigen value i.e., $\lambda = \lambda_n = -1 - n^2\pi^2$ then the given integral equation does not possess any solution.

Example: Solve the symmetric integral equation in the form
 $\phi(x) = 1 + \lambda \int_0^\pi \cos(x + \xi) \phi(\xi) d\xi$

Solution: Considering the corresponding homogeneous integral equation in the form

$$\phi(x) = \lambda \int_0^\pi \cos(x + \xi) \phi(\xi) d\xi \quad (8)$$

The required eigen values are $\lambda_1 = \frac{2}{\pi}$ and $\lambda_2 = -\frac{2}{\pi}$

The eigen function corresponding to the eigen value $\lambda_1 = \frac{2}{\pi}$ is

$$\phi_1(x) = \frac{2\mu}{\pi} \cos x = \cos x, \text{ let } \left(\frac{2\mu}{\pi}\right) = 1$$

The corresponding normalized eigen function $\bar{\Phi}_1(x)$ is determined by

$$\bar{\Phi}_1(x) = \frac{\phi_1(x)}{\left\{\int_0^\pi [\phi_1(x)]^2 dx\right\}^{1/2}} = \frac{\cos x}{\left\{\int_0^\pi \cos^2 x dx\right\}^{1/2}} = \sqrt{\frac{2}{\pi}} \cos x \quad (9)$$

Also, the eigen function corresponding to the eigen value $\lambda_2 = -\frac{2}{\pi}$ is

$$\phi_2(x) = \frac{2\nu}{\pi} \sin x = \sin x, \text{ Let } \left(\frac{2\nu}{\pi}\right) = 1$$

The corresponding normalized eigen function $\Phi_2(x)$ is determined by

$$\Phi_2(x) = \frac{\phi_2(x)}{\left\{\int_0^\pi [\phi_2(x)]^2 dx\right\}^{1/2}} = \frac{\sin x}{\left\{\int_0^\pi \sin^2 x dx\right\}^{1/2}} = \sqrt{\frac{2}{\pi}} \sin x \quad (10)$$

$$\text{Also } F_1(x) = \int_0^\pi F(x) \Phi_1(x) dx = \int_0^\pi \sqrt{\frac{2}{\pi}} \cos x dx = 0$$

$$F_2(x) = \int_0^\pi F(x) \Phi_2(x) dx = \int_0^\pi \sqrt{\frac{2}{\pi}} \sin x dx = 2\sqrt{\frac{2}{\pi}}$$

Put Your Own Notes

- (i) The integral will possess unique solution which will be given by (when $\lambda \neq \lambda_1, \lambda \neq \lambda_2$)

$$\phi(x) = F(x) + \lambda \sum_{m=1}^{\infty} \frac{f_m}{\lambda_m - \lambda} \phi_m(x).$$

$$\text{or } \phi(x) = F(x) + \frac{\lambda}{\lambda_1 - \lambda} F_1 \phi_1(x) + \frac{\lambda}{\lambda_2 - \lambda} F_2 \phi_2(x)$$

$$\text{or } \phi(x) = 1 + \frac{\lambda}{-\left(\frac{2}{\pi}\right) - \lambda} 2\sqrt{\frac{2}{\pi}} \cdot \sqrt{\frac{2}{\pi}} \sin x = 1 - \frac{4\lambda \sin x}{2 + \lambda\pi}$$

- (ii) When $\lambda = \lambda_2 = -\left(\frac{2}{\pi}\right)$, $F_2(x) \neq 0$ then the integral equation possesses no solution.

- (iii) When $\lambda = \lambda_1 = \left(\frac{2}{\pi}\right)$, $F_1(x) \neq 0$ then $\exists$ infinite many solutions given by

$$\phi(x) = F(x) + C_1 \Phi_1(x) + \frac{\lambda}{\lambda_2 - \lambda} F_2 \Phi_2(x)$$

$$\text{or } \phi(x) = 1 + C_1 \left(\frac{2}{\pi}\right)^{1/2} \cos x + \frac{\left(\frac{2}{\pi}\right)}{-\left(\frac{2}{\pi}\right) - \left(\frac{2}{\pi}\right)} 2\sqrt{\frac{2}{\pi}} \cdot \sqrt{\frac{2}{\pi}} \sin x$$

$$\text{or } \phi(x) = 1 + C \cos x - \sqrt{\frac{2}{\pi}} \sin x, \text{ where } C = C_1 \sqrt{\frac{2}{\pi}}.$$

EXERCISE - 4

1. Solve the integral equation

$$\phi(x) = x + \lambda \int_0^1 K(x, \xi) \phi(\xi) d\xi$$

$$K(x, \xi) = \begin{cases} x(\xi - 1) & 0 \leq x \leq \xi \\ \xi(x - 1) & \xi \leq x \leq 1 \end{cases}$$

2. Using Hilbert-Schmidt theorem, solve the symmetric integral equation

$$(a) \phi(x) = (x + 1)^2 + \int_{-1}^1 (x\xi + x^2\xi^2) \phi(\xi) d\xi$$

$$(b) \phi(x) = (x^2 + 1) + \frac{3}{2} \int_{-1}^1 (x\xi + x^2\xi^2) \phi(\xi) d\xi$$

$$(c) \phi(x) = x + \lambda \int_0^{2\pi} \sin(x + \xi) \phi(\xi) d\xi$$

3. Solve the integral equation

$$\phi(x) = e^x + \lambda \int_0^1 (5x^2 - 3) \xi^2 \phi(\xi) d\xi$$

ANSWERS EXERCISE (1)

Put Your Own Notes

1. (a) $\phi(x) = -1 - \int_0^x (2x - \xi)\phi(\xi)d\xi$
2. (a) $\phi(x) = c_1\left(3 + 6x - \frac{5}{2}x^2\right) + c_2(6 - 5x) - 5c_3 + \int_0^x \left[3 + 6(x - \xi) - \frac{5}{2}(x - \xi)^2\right]\phi(\xi)d\xi$
- (b) $\phi(x) = \cos x - 2x\left(1 + x^2\right) - \int_0^x (1 + x^2)(x - \xi)\phi(\xi)d\xi$
- (c) $\phi(x) = \left(1 - x + \frac{1}{6}x^3\right) + \int_0^x \left[\sin \xi - (x - \xi)(e^\xi + \cos \xi)\right]\phi(\xi)d\xi$
- (d) $\phi(x) = \left(\frac{1}{6}\right)(x^3 - 3x) + \lambda \int_0^1 K(x, \xi)\phi(\xi)d\xi$

$$\text{Where } K(x, \xi) = \begin{cases} x; & x < \xi \\ \xi; & x > \xi \end{cases}$$

ANSWERS EXERCISE (2)

1. (a) $e^{(1+\lambda)(x-\xi)}, x > \xi \forall \lambda$
- (b) $\frac{2 + \cos x}{2 + \cos \xi} e^{\lambda(x-\xi)}$
- (c) $(x - \xi + 2)e^{x-\xi}$
- (d) $2xe^{x^2-\xi^2}$
2. (a) $-\cos x = \sin x - \frac{1}{2}x \sin x$
- (b) $\phi(x) = 93e^{3x} - 64e^{2x}$
- (c) $e^x(1+x^2)$
- (d) $1 + \frac{x^3}{2} + \frac{x^6}{2.5} + \frac{x^9}{2.5.8} + \dots$
3. (a) $\sin x$
- (b) $\cos hx$
- (c) e^x
- (d) e^x
4. (a) $K_{2v-1}(x, \xi) = (2\pi)^{2v-2}(x + \sin \xi)$
and $K_{2v}(x, \xi) = (2\pi)^{2v-1}(1 + x \sin \xi)$ when $v = 1, 2, 3, \dots$
- (b) $K_v(x, \xi) = (-1)^{v-1} \left(\frac{1+e^\pi}{2}\right)^{v-1} e^x \cos \xi, v = 1, 2, \dots$
5. (a) $\phi(x) = \frac{3}{2}e^x - \frac{1}{2}xe^x - \frac{1}{3}e + 1$
- (b) $\phi(x) = x$
- (c) $\phi(x) = e^x$
- (d) $\phi(x) = 1 + \lambda + \left(x + \frac{1}{2}\right) + \lambda^2 \left(x + \frac{7}{12}\right) + \lambda^3 \left(\frac{13}{12}x + \frac{5}{8}\right)$

ANSWERS EXERCISE (3)

Put Your Own Notes

1. (a) $R(x, \xi; \lambda) = \frac{(2x - \xi) + \lambda \left(2x + \xi - 2\xi x - \frac{2}{3} \right)}{1 - \frac{\lambda}{4} + \frac{\lambda^2}{6}}$
- (b) $R(x, \xi; \lambda) = \frac{x^2 \xi - x \xi^2 + x \xi + \left(\frac{x + \xi}{4} - \frac{x \xi}{3} - \frac{1}{5} \right)}{1 + \frac{\lambda^2}{240}}$
- (c) $R(x, \xi; \lambda) = \frac{(\sin x - \sin \xi) - \pi(1 + 2 \sin x \sin \xi)}{1 + 2\lambda^2 \pi^2}$
- (d) $R(x, \xi; \lambda) = \frac{1 + 3x\xi + \lambda \left[\frac{\{3(x + \xi)\}}{2} - 3x\xi - 1 \right]}{1 - 2\lambda + \left(\frac{\lambda}{4} \right) \lambda^2}$
2. (a) $\phi(x) = \sec^2 x + \frac{\lambda}{1 - \lambda} \tan 1$
- (b) $\phi(x) = e^x + \frac{\lambda x}{1 - \left(\frac{10^3}{3} \right) \lambda} [9e^{10} + 1]$
- (c) $\phi(x) = \sin x + \frac{2\lambda x}{1 - 42\lambda} \sin 7 \sin 3$
3. (a) $\phi(x) = \cos x$
- (b) $\phi(x) = 2x - \pi + \frac{\pi^2 \sin x}{\pi - 1}$
- (c) $\phi(x) = x + \frac{\lambda}{(12 - 24\lambda - \lambda^2)} [10 + (6 + \lambda)x]$
- (d) $\phi(x) = \left[\frac{(240 - 60\lambda)x + 80\lambda x^2}{240 - 120\lambda - \lambda^2} \right]$
- (e) $\phi(x) = \frac{6}{5}(1 - 4x) + \frac{48}{48 + 29\lambda^2} \left[2\lambda^2 x + \left(\frac{\lambda + \lambda^2}{4} \right) \log x \right]$
4. (a) Eigen value $\lambda = \frac{1}{4}$, corresponding eigen function $\phi(x) = x^2 + \frac{3x}{2}$
- (b) eigen values $\lambda_1 = \lambda_2 = -3$, corresponding eigen function $\phi(x) = x - 2x^2$
- (c) Equation has trivial solution, so it does not contain any eigen values or eigen functions

ANSWERS EXERCISE (4)

Put Your Own Notes

1. $\phi(x) = x + \frac{2\lambda}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n \sin n\pi x}{n(n^2\pi^2 + \lambda)}$; $\lambda \neq -n^2\pi^2$, $n=1,2,\dots$

2. (a) $\phi(x) = \frac{25}{9}x^2 + 6x + 1$

(b) $\phi(x) = 5x^2 + Ax + 1$, where $A = \frac{(C\sqrt{6})}{2}$

(c) $\phi(x) = x - \frac{2\lambda\pi\sqrt{\pi}}{1-\lambda^2\pi^2} (\lambda\pi \sin x + \cos x)$

ASSIGNMENT SHEET

- The integral equation
$$g(x)y(x) = f(x) + \lambda \int_{\alpha}^{\beta} k(x,t)y(t)dt$$
 with $f(x)$, $g(x)$ and $k(x,t)$ as known functions, α and β as known constant, and λ as a known parameter, is a
 - Linear integral equation of Volterra type
 - Linear integral equation of Fredholm type
 - Non-linear integral equation of Volterra type
 - Non-linear integral equation of Fredholm type
- Let $y(x) = f(x) + \lambda \int_a^b k(x,t)y(t)dt$, where $f(x)$ and $k(x,t)$ are known functions a and b are known constants and λ is a known parameter. If λ be the Eigen values of the corresponding homogenous equation, then the above integral equation has in general,
 - Many solution for $\lambda \neq \lambda_i$
 - No solution for $\lambda \neq \lambda_i$
 - A unique solution for $\lambda = \lambda_i$
 - Either many solutions or no solution at all for $\lambda = \lambda_i$, depending on the form of $f(x)$
- Consider the integral equation
$$y(x) = x^2 + \lambda \int_0^1 xty(t)dt$$
, where λ is a real parameter. Then the Neumann series for the integral equation converges for all values of λ
 - Except for $\lambda = 3$
 - Lying in the interval $-3 < \lambda < 0$
 - Lying in the interval $-3 < \lambda < 3$
 - Lying in the interval $0 < \lambda < 3$
- The solution of the integral equation
$$\phi(x) = \frac{5x}{6} + \frac{1}{2} \int_0^1 xt\phi(t)dt$$
 satisfies
 - $\phi(0) + \phi(1) = 1$
 - $\phi\left(\frac{1}{2}\right) + \phi\left(\frac{1}{3}\right) = 1$
 - $\phi\left(\frac{1}{4}\right) + \phi\left(\frac{1}{2}\right) = 1$
 - $\phi\left(\frac{3}{4}\right) + \phi\left(\frac{1}{4}\right) = 1$
- A solution of the integral equation
$$\phi(x) = \int_0^x e^{x-t}\phi(t)dt + \sinh x$$
 is
 - $\phi(x) = e^{-x}$
 - $\phi(x) = e^x$
 - $\phi(x) = \sinh x$
 - $\phi(x) = \cosh x$
- Which of the following is a degenerate kernel?
 - $k(x,t) = \sum_{n=1}^{\infty} x^n t^n$
 - $k(x,t) = e^{|x|+t}$
 - $k(x,t) = e^{|x-t|}$
 - $k(x,t) = e^{xt}$
- A solution of the integral equation
$$\int_0^x 3^{x-t}\phi(t)dt = x$$
 is
 - $\phi(x) = 3x^e$
 - $\phi(x) = 1 - 3x^e$
 - $\phi(x) = 1 - x \log 3$
 - $\phi(x) = x \log 3$

8. The equation $u(x) = f(x) + \int_a^x K(x, t)u(t) dt$ is
- A Volterra's linear integral equation of the first kind.
 - A Fredholm's linear integral equation of the first kind
 - A Volterra's linear integral equation of the second kind
 - A Fredholm's linear integral equation of the second kind.

9. The integral equation corresponding to the initial value problem

$$\frac{d^2 y}{dx^2} + y = 0, y(0) = 0, y'(0) = 1 \text{ is}$$

- $u(x) = x + \int_0^1 (t-x)u(t) dt$
- $u(x) = x + \int_0^x (t-x)u(t) dt$
- $u(x) = \sin x + \int_0^x (t-x)u(t) dt$
- $u(x) = x(1+x) + \int_0^1 (t-x)u(t) dt$

10. The linear integral equation associated with the boundary value problem

$$\frac{d^2 u}{dx^2} + \lambda u = 0, u(0) = 0, u(1) = 0 \text{ is}$$

- A Fredholm linear integral equation of the first kind
- A Volterra linear integral equation of the first kind
- A Fredholm linear integral equation of the second kind
- A Volterra linear integral equation of the second kind.

11. For the Volterra type linear integral equation

$$\phi(x) = x + 2 \int_0^x e^{x-\zeta} \phi(\zeta) d\zeta, \text{ the resolvent}$$

kernel $R(x, \zeta; 2)$ of the kernel $e^{x-\zeta}$ is

- $(e-\zeta)^2 e^{2(x-\zeta)}$
- $(e-\zeta) e^{x-\zeta}$
- $e^{3(x-\zeta)}$
- $e^{(x-\zeta)}$

12. The integral equation, involving a parameter

$$\lambda, \phi(x) = \cos x + \lambda \int_0^\pi \cos(x+t) \phi(t) dt \text{ has}$$

- A unique solution if $\lambda = 1$, and an infinite number of solution if $\lambda = \frac{2}{\pi}$
- A unique solution if $\lambda = -1$, and an infinite number of solution if $\lambda = -\frac{2}{\pi}$

- A unique solution if $\lambda \neq \frac{2}{\pi}$

- No solution if $\lambda = \pm \frac{2}{\pi}$

13. For the integral equation

$$u(x) = f(x) + \lambda \int_a^b K(x, t)u(t) dt \text{ to have a}$$

continuous solution in the interval $a \leq x \leq b$, which of the following assumptions are necessary?

- $K(x, t) \neq 0$, is real and continuous in the region $a \leq x \leq b, a \leq t \leq b$ with $|K(x, t)| \leq M$

- $f(x) \neq 0$, is real and continuous in the interval $a \leq x \leq b$,

- λ is a constant

- $|\lambda| < \frac{1}{M(b-a)}$

14. The integral equation

$$y(x) = 1 + \lambda \int_0^{\pi/2} \cos(x-t) y(t) dt \text{ has}$$

- A unique solution for $\lambda \neq 4/(\pi+2)$

- A unique solution for $\lambda \neq 4/(\pi-2)$

- No solution for $\lambda \neq 4/(\pi+2)$, but the corresponding homogeneous equation has non-trivial solutions

- No solution for $\lambda \neq 4/(\pi-2)$, but the corresponding homogenous equation has non-trivial solutions.

15. For the linear integral equation

$$\phi(x) = x + \int_0^{1/2} \phi(\xi) d\xi, \text{ the resolvent kernel}$$

$R(x, \xi; 1)$ is

- (a.) $\frac{1}{2}$
(b.) 2
(c.) $\frac{3}{2}$
(d.) 4

16. The integral equation

$$\phi(x) = \lambda \int_0^\pi k(x, \zeta) \phi(\zeta) d\zeta \quad \text{Where } \lambda \text{ is a}$$

parameter,

and

$$K(x, \zeta) = \begin{cases} \cos \zeta \sin x, & \text{for } 0 \leq \zeta \leq x \\ \sin \zeta \cos x, & \text{for } x \leq \zeta \leq \pi \end{cases} \text{ leads to a}$$

boundary value problem

$$\phi''(x) - f(\lambda)\phi(x) = 0, \phi(\pi) = 0, \phi'(0) = 0$$

where $f(\lambda)$ is known. Then the boundary value problem has.

- (a.) a unique solution when $f(\lambda) = 0$
(b.) infinite number of solution when $f(\lambda) > 0$
(c.) no solution when $f(\lambda) < 0$
(d.) a unique solution when $\lambda > 1$

17. For the homogenous Fredholm integral

$$\text{equation } \phi(x) = \lambda \int_0^1 e^{x+t} \phi(t) dt, \text{ a non-trivial}$$

solution exists, then λ has the value

- (a.) $\lambda = \frac{2}{e-1}$
(b.) $\lambda = \frac{1}{e^2+1}$
(c.) $\lambda = \frac{1}{e+1}$
(d.) $\lambda = \frac{2}{e^2-1}$

18. The initial value problem $\frac{d^2 y}{dx^2} + y = 0; x > 0$
 $y(0) = -1, y'(0) = 0$, is equivalent to the Volterra integral equation

(a.) $y(x) = 1 + \int_0^x (t-x)y(t) dt$

(b.) $y(x) = 1 + \int_0^x (t+x)y(t) dt$

(c.) $y(x) = 1 + \int_0^x xy(t) dt$

(d.) $y(x) = 1 + \int_0^x (x-t)y(t) dt$

19. The integral equation

$$\phi(x) - \lambda \int_{-1}^1 \cos[\pi(x-t)] \phi(t) dt = f(x) \text{ has}$$

- (a.) A unique solution for $\lambda \neq 1$ when $f(x) = x$
(b.) No solution for $\lambda \neq 1$ when $f(x) = 1$
(c.) No solution for $\lambda = 1$ when $f(x) = x$
(d.) Infinite number of solutions for $\lambda = 1$ when $f(x) = 1$.

20. The solution of the integral equation

$$g(s) = s + \int_0^1 su^2 g(u) du \text{ is given by}$$

(a.) $g(t) = \frac{3t}{4}$

(b.) $g(t) = \frac{4t}{3}$

(c.) $g(t) = \frac{2t}{3}$

(d.) $g(t) = \frac{3t}{2}$

21. If $y(t) = 1 + \int_0^t y(v) e^{-(t+v)} dv$ then $y(t)$ at $t=1$ equals

- (a.) 0
(b.) 1
(c.) 2
(d.) 3

22. The initial value problem corresponding to the integral equation $y(x) = 1 + \int_0^x y(t) dt$ is

- (a.) $y' - y = 0, y(0) = 1$
- (b.) $y' + y = 0, y(0) = 0$
- (c.) $y' - y = 0, y(0) = 0$
- (d.) $y' + y = 0, y(0) = 1$

23. Let $k(x, t) = \begin{cases} x+t & 0 \leq t \leq x \\ 0 & \text{otherwise} \end{cases}$

Then, the integral equation

$$y(x) = 1 + \lambda \int_0^1 y(t) k(x, t) dt \text{ has}$$

- (a.) A unique solution for every value of λ
- (b.) No solution for any value of λ
- (c.) A unique solution for finitely many values of λ only
- (d.) Infinitely many solutions for finitely many values of λ

24. The integral equation

$$y(x) = \int_0^x (x-t)y(t) dt - x \int_0^1 (1-t)y(t) dt \text{ is}$$

equivalent to

- (a.) $y'' - y = 0, y(0) = 0, y(1) = 0$
- (b.) $y'' - y = 0, y(0) = 0, y'(0) = 0$
- (c.) $y'' + y = 0, y(0) = 0, y(1) = 0$
- (d.) $y'' + y = 0, y(0) = 0, y'(0) = 0$

25. The integral equation

$$y(x) = \lambda \int_0^{2\pi} \sin(x+t)y(t) dt \text{ has}$$

- (a.) No solution for any value of λ
- (b.) Unique solution for every value of λ
- (c.) Infinitely many solutions for only one value of λ
- (d.) Infinitely many solutions for two values λ

26. Given that the eigen values of the integral equation $y(x) = \lambda \int_0^{2\pi} \cos(x+t)y(t) dt$ are $\frac{1}{\pi}$

and $-\frac{1}{\pi}$ with respective eigen functions $\cos x$ and $\sin x$. Then the integral equation $y(x) = \sin x + \cos x + \lambda \int_0^{2\pi} \cos(x+t)y(t) dt$ has

- (a.) Unique solution for $\lambda = 1/\pi$
- (b.) Unique solution for $\lambda = -1/\pi$
- (c.) Unique solution for $\lambda = \pi$
- (d.) No solution for $\lambda = -\pi$

27. The values of λ for which the integral equation $y(x) = \lambda \int_0^1 (6x-t)y(t) dt$ Has a non trivial solution, are given by the roots of the equation

- (a.) $(3\lambda - 1)(2 + \lambda) - \lambda^2 = 0$
- (b.) $(3\lambda - 1)(2 + \lambda) + 2 = 0$
- (c.) $(3\lambda - 1)(2 + \lambda) - 4\lambda^2 = 0$
- (d.) $(3\lambda - 1)(2 + \lambda) + \lambda^3 = 0$

28. The eigen values λ of the integral equation

$$y(x) = \lambda \int_0^{2\pi} \sin(x+t)y(t) dt \text{ are}$$

- (a.) $\frac{1}{2\pi}, -\frac{1}{2\pi}$
- (b.) $\frac{1}{\pi}, -\frac{1}{\pi}$
- (c.) $\pi, -\pi$
- (d.) $2\pi, -2\pi$

29. The integral equation

$$y(x) = x - \int_0^x (x-t)y(t) dt \text{ is solved by the}$$

method of successive approximations. Starting with initial approximation $y_1(x) = x$ the second approximation $y_2(x)$ is given by

- (a.) $y_2(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!}$
- (b.) $y_2(x) = x + \frac{x}{3!}$
- (c.) $y_2(x) = x - \frac{x^3}{3!}$
- (d.) $y_2(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!}$

30. Solution of the initial value problem

$$\frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_2(x)y = F(x), \quad 0 \leq x \leq 1$$

$$y(0) = c_0, \left(\frac{dy}{dx} = x \right)_{x=0} = c_1, \quad \text{Where}$$

$a_1(x), a_2(x)$ and $F(x)$ are continuous functions on $\{0,1\}$, may be reduced, in general to a solution of some linear

- (a.) Fredholm integral equation of first kind
- (b.) Volterra's integral equation of first kind
- (c.) Fredholm integral equation of second kind
- (d.) Volterra's integral equation of second kind

31. Which of the following functions is a solution of the Volterra type integral equation

$$f(x) = x + \int_0^x (\sin(x-t)f(t))dt$$

- (a.) $x + \frac{x^3}{3}$
- (b.) $x - \frac{x^3}{3}$
- (c.) $x + \frac{x^3}{6}$
- (d.) $x - \frac{x^3}{6}$

32. Which of the following functions is a solution of the Fredholm type equation

$$f(x) = x + \int_0^1 (xtf(t))dt$$

- (a.) $2x/3$
- (b.) $3x/2$
- (c.) $3x/4$
- (d.) $4x/3$

33. Suppose

$$y(x) = \lambda \int_0^{2\pi} y(t) \sin(x+t) dt, \quad x \in [0, 2\pi]$$
 has

eigenvalue $\lambda = \frac{1}{\pi}$ and $\lambda = -\frac{1}{\pi}$, with

corresponding eigen functions

$$y_1(x) = \sin(x) + \cos(x) \quad \text{and}$$

$$y_2(x) = \sin(x) - \cos(x), \text{ respectively. Then}$$

the integral equation

$$y(x) = f(x) + \frac{1}{\pi} \int_0^{2\pi} y(t) \sin(x+t) dt, \quad x \in [0, 2\pi]$$

has a solution when $f(x)$ is

- (a.) 1
- (b.) $\cos(x)$
- (c.) $\sin(x)$
- (d.) $1 + \sin(x) + \cos(x)$

34. The value of α for which the integral equation $u(x) = \alpha \int_0^1 e^{x-t} u(t) dt$, has a non-trivial solution is

- (a.) -2
- (b.) -1
- (c.) 1
- (d.) 2

35. The integral equation

$$x(t) = \sin t + \lambda \int_0^1 (s^2 t^3 + e^{x^2+t^2}) x(s) ds$$

$0 \leq t \leq 1, \lambda \in R, \lambda \neq 0$ has a solution for

- (a.) All non-zero values of λ
- (b.) No value of λ
- (c.) Only countably many positive values of λ
- (d.) Only countably many negative values of λ

36. The integral equation

$$x(t) - \int_0^1 [\cos t \sec s x(s)] ds = \sin ht, \quad 0 \leq t \leq 1,$$

has

- (a.) No solution
- (b.) A unique solution
- (c.) More than one but finitely many solution
- (d.) Infinitely many solutions

37. The resolvent kernel for the integral equation

$$u(x) = F(x) + \int_{\log 2}^x e^{(t-x)} u(t) dt \text{ is}$$

- (a.) $\cos(x-t)$
- (b.) 1
- (c.) e^{t-x}
- (d.) $e^{2(t-x)}$

38. For a continuous function $f(t)$, $0 \leq t \leq 1$, the

$$\text{integral equation } y(t) = f(t) + 3 \int_0^1 t s y(s) ds$$

has

- (a.) a unique solution if $\int_0^1 s f(s) ds \neq 0$
- (b.) no solution if $\int_0^1 s f(s) ds = 0$
- (c.) infinitely many solution if $\int_0^1 s f(s) ds = 0$
- (d.) infinitely many solutions if $\int_0^1 s f(s) ds \neq 0$

39. The eigen value λ of the following Fredholm integral equation

$$y(x) = \lambda \int_0^1 x^2 t y(t) dt, \text{ is}$$

- (a.) -2
- (b.) 2
- (c.) 4
- (d.) -4

40. The function $\phi(x)$ satisfying the integral

$$\text{equation } \int_0^x e^{x-\xi} \phi(\xi) d\xi = \frac{x^2}{2} \text{ is}$$

- (a.) $\frac{x^2}{2}$
- (b.) $x + \frac{x^2}{2}$
- (c.) $x - \frac{x^2}{2}$
- (d.) $1 + \frac{x^2}{2}$

41. The solution of this integral equation Consider the Fredholm integral equation

$$u(x) = x + \lambda \int_0^1 x e^t u(t) dt \text{ is}$$

- (a.) $\frac{x+1}{1-\lambda}$
- (b.) $\frac{x^2}{1-\lambda^2}$
- (c.) $\frac{x}{1+\lambda^2}$
- (d.) $\frac{x}{1-\lambda}$

42. The resolvent kernel $R(x, t; \lambda)$ for this integral equation is

- (a.) $\frac{x e^t}{1-\lambda}$
- (b.) $\frac{\lambda x e^t}{1+\lambda}$
- (c.) $\frac{x e^t}{1+\lambda^2}$
- (d.) $\frac{x e^t}{1-\lambda^2}$